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Optimization and lubrication properties of Malaysian crude palm oil fatty acids based neopentyl glycol diester green biolubricant

Nurazira Mohd Nor [a,*], Nadia Salih [b], Jumat Salimon [b,**]

[a] Material, Inorganic and Oleochemistry (MaterOleo) Research Group, Faculty of Applied Sciences, Universiti Teknologi MARA Cawangan Negeri Sembilan, Kampus Kuala Pilah, Pekan Parit Tinggi, 72000 Kuala Pilah, Negeri Sembilan, Malaysia
[b] Department of Chemical Sciences, Faculty of Science and Technology, Universiti Kebangsaan Malaysia, 43600 Bangi, Selangor, Malaysia

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ABSTRACT

Green biolubricant base stock based on crude palm oil fatty acids (CPOFAs) was produced through acid catalyst esterification of palm oil fatty acids with neopentyl glycol (NPG). The response surface methodology (RSM) of D-optimal design was employed to optimize the esterification process. The resultant polyol ester, crude palm oil fatty acids-NPG diester (CPOFAs-NPGDE) was evaluated its lubrication properties. The optimal condition for the esterification process was obtained at CPOFAs:NPG molar ratio of 2.26:1, 1.12% H_2SO_4 catalyst, reaction temperature of 138 °C for 4.79 h reaction time, respectively. The optimal CPOFAs-NPGDE produced was 87.6% yield with 100% diester selectivity. The resultant ester showed good lubrication properties with oxidative stability temperature at 184 °C, pour point at 10 °C, flash point at 235 °C and viscosity index of 190. Tribological tests showed that COFAs-NPGDE has shown low coefficient friction of hydrodynamic regime of 0.20 at 40 °C and 0.19 at 100 °C and comparable with some commercial lubricants. Rheological tests show that CPOFAs-NPGDE can be classified as a Newtonian fluid with good lubrication properties within ISO VG 46 lubricant grade. This make it plausible to be used for green industrial lubrication applications such as hydraulic oil, compressor and turbine fluids.

1. Introduction

Malaysia is one of the world's largest palm oils (PO) exporters and second largest production after Indonesia [1,2]. The country dominates 27% of the oils and fats trade and 11% of oils and fats production globally. Palm oil fresh fruit bunches (FFBs) produce about 25% crude palm oil (CPO) or 65% from the flesh (mesocarp) of its fruit and crude palm kernel oil (CPKO) from the seed or kernel [3]. Over the years, CPO has been used in several important ingredients for food and non-food applications. Many uses of CPO are due to almost equal amounts of saturated (0.5% myristic, 41.8% palmitic and 3.7% stearic) and unsaturated (44.4% oleic and 9.7% linoleic) fatty acids composition [4]. It serves many uses of non-economic grade CPO such as high free fatty acids CPO (HFFA-CPO) for non-food products such as biolubricant [5]. HFFA-CPO normally produced during the FFBs harvesting period of raining or monsoon season or under lack of supervision manners such as unripe or over ripe FFBs. The adverse impacts of fossil fuel, petroleum and petroleum-based materials such as lubricants, have pushed researchers and industries interest toward plant oils as alternative raw material resources. In specific, many works are now focusing on modifying the chemical properties of plant oils to make it suitable for replacing non-environmentally friendly conventional fossil fuel-derived lubricants [6]. Consequently, HFFA-CPO is encouraged become major raw materials for numerous oleochemical products industries, including the production of biolubricants, biodiesel and bioplastics [4]. With the advancements in oleochemicals technology nowadays and uneconomical reason, HFFA-CPO are increasingly demanded as feed stock for green biolubricants production [7].

Due to escalating alarms over the influence of petroleum oil-based fluids on environment, is generating sufficient prospects for bio-based lubricants (biolubricants). Biolubricants global market estimated at US $1.9 billion in the year 2020 is projected to reach a revised size of US $2.5 billion by 2026 with 5.2% Compound Annual Grow Rate (CAGR). Highest market was the U.S. Market at $360.2 million in 2021, while China market is forecast to reach $230.4 million by 2026. Japan and Canada each are forecasted to grow at 3.7% and 4.1% CAGR

* Corresponding author.
** Corresponding author.
E-mail addresses: nurazira@uitm.edu.my (N.M. Nor), nadiaalnami@hotmail.com (N. Salih), jumatsal@gmail.com (J. Salimon).

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respectively. Within Europe, Germany is forecast to grow at approximately 5% CAGR while rest of European market will reach US$246.8 million by the end of 2026. The drive demand for biolubricants is due to the advancements in the engine design development and operating efficiency, strict government regulations particularly in North America and Europe, that discourage the use of synthetic and chemical-based lubricants. Fast exhausting crude oil reserves and crude oil produce also initiate high demand for industrial biolubricants. In the European region, subsidies, tax breaks, and labelling programs triggers high demand for biolubricants, particularly for the hydraulic fluid application. It transmits the power in hydraulic machineries which are predominantly used in mining, industrial, agricultural and marine sectors. In this market region alone is expected to reach $884.6 million by 2026 [8].

Green biolubricants should have excellent lubricating properties, high flash points and high viscosity indices. However, the low oxidative stability of plant oils (triacylglycerols), as well as their poor properties at low temperature become significant barriers hindering their take-up as industrial biolubricants [9]. Since plant oil tri-esters consist of β-H atom on the middle carbon of glycerol in its triacylglycerol backbone, it has low hydrolytic and thermo-oxidative stabilities and poorly flow property at cold conditions. This makes plant oil ester more susceptible to the elimination reaction leading to the subsequent degradation of the molecule, which is thermally unstable [9,10]. Therefore, it is not suitable for direct application as biolubricants and this drawback need to be mitigated through green modification process. Green biolubricants are recent emerging lubricants derived from plant oils and fats. These green biolubricants are biodegradable, renewable and environmentally friendly [11]. There are intense focuses have been directed towards exploring the industrial application potential of green biolubricants [12]. Moreover, various industrial products are now produced from plant oil-based fatty acids, in particular low quality HFFA-CPO. Since CPO has been produced in large amount annually, nearly 81 million matric tones annually [13], crude palm oil fatty acids could be produced in extensive amount readily available for industrial usage to produce high-end products. These crude palm oil fatty acids (CPOFAs) were obtained from the hydrolysis and separation from CPO which is cheap and natural abundantly produce as compared to palm industry by-products of palm fatty acids distillate (PFAD). Since CPOFAs is having nearly the same chemicals composition properties, it is expected having the same industrial usages as PFAD. Therefore, CPOFAs is more reliable and sustainable as industrial green renewable raw materials base stock [10] for their specific products production such as laundry soaps [14], biodiesel [15], bioplastics [16], advanced fuels and chemicals [17], biofuels [18] and green biolubricants [12,19].

Various industrial modifications were achieved through simple or mid-complexes chemicals or bio-transformation to produce their targeted products. Several chemical modifications processes such as transesterification, esterification, epoxidation, epoxide ring opening are among important industrial process due to their simplicity, viable, achievable and low-cost processes. The processes have been reported elsewhere especially in the production of esters-based biolubrcants [12]. The esterification of plants oil-based fatty acids with various polyhydric alcohols for example, have been reported successfully achieved with high yields percentage and high esters selectivity. Neopentyl glycol (NPG) [19], trimethylolpropane (TMP) [7] and pentaerythritol (PE) [11] are examples of polyhydric alcohols used [12]. Polyol ester base stocks are commercially synthesized and engineered to deliver the highest performance lubrication under the most demanding conditions. Polyol esters exhibit extraordinary stability (thermally, oxidation, hydrolytic) due to the absence of hydrogen in the beta position (β-hydrogen) as well as the presence of a central quaternary carbon [11]. Thus, the lubrication properties are enhancing [12] especially towards thermal [20] and hydrolytic [21] oxidative stability at high temperature as compared to natural esters. Several reports have highlighted biolubricant based on NPG with fatty acids, plants biodiesel, plant oil processing by-product or microbial oil through which are suitable for many industrial, mining and aviation application [22].

NPG ester for use as friendly biolubricants has been synthesized either by microbial enzymatic or chemicals catalysts. Enzymatic production of NPG esters catalyzed by lipase B from *Candida antarctica* (Novozyme 435) and from *Rhizomucor miehei* (Lipozyme RM IM) has been reported. In solvent-free medium, Lipozyme RM IM maintained high conversions (>95%) from soybean fatty acid methyl esters in 24 h whereas from castor fatty acid methyl esters requires 48 h were obtained [23]. Biocatalysts lipase from *Candida rugosa* (CRL) sustained high conversions of about 80–90% of NPG and TMP esters for biolubricants derived from soybean fatty acids distillate (SFAD) after 3–6 h [24]. Extracted microbial oils from confectionery and wheat milling side streams fermentation were enzymatically hydrolyzed and the free fatty acids were esterified by Lipomod 34-MDP in a solvent-free system with trimethylolpropane (TMP) and neopentyl glycol (NPG). The highest conversion yields were 88% and 82.7% for NPG esters of *Rhodosporidium toruloides* and *Cryptococcus curvatus*, respectively [25]. Palm-based NPG diester as a potential green insulating oil has been synthesized via transesterification of high oleic palm oil methyl ester (HOPME) with NPG. The optimum conditions inferred from the analyses were: molar ratio of HOPME to NPG of 2:1.3, temperature of 182 °C, pressure of 0.6 mbar for 1 h and catalyst concentration of 1.2% to produce the maximum NPG diester yield of 87 wt% [26]. NPG dioleate and NPG diundecenoate esters as lubrication oil on machine and spindle bearing textile industries have been synthesized by optimal esterification reaction conditions: molar ratio 1:2.5 (alcohol, acid), *p*-TSA catalyst 1% wt, temperature 120–130 °C. Over 89% of conversion was obtained in 4–5 h. These synthesized esters meet the standard ISO VG 10 and 12 grade properties [27]. Hexanoic acid (HA)-based NPG ester was optimized using Taguchi L9 experimental layout. The optimum reaction conditions are identified to be 423 K, 0.5 kPa; sulfuric acid catalyst 0.5% w/w and HA:NPG molar ratio at 3:1. At this optimum condition the neopentyl glycol conversion obtained was 99% [28]. 100% of NPG dioleate was successfully produced by esterification of NPG and oleic acid. The optimized condition for esterification of NPG and oleic acid was obtained at molar ratio of NPG:oleic acid of 1:2.3, temperature of 130 °C and catalyst amount of 2% for 4 h [29]. However so far, no single study regarding biolubricant based on NPG and low-cost materials of crude palm oil fatty acids has been reported. Therefore, it attracts many researchers to explore the modifications of crude palm oil and their process optimization to tap its great industrial beneficials. Due to its available in large quantities, neopentyl glycol also serves as low-cost reactant as compared to other polyhydric alcohols for the production of biolubricants. It provides more environmentally friendly reaction conditions with high yield due to less molecular stearic hindrance as compared to TMP or PE. Furthermore, no effects on aquatic organisms were observed in the available studies for the NPG esters on any of the three trophic levels (fish, daphnia, algae) up to the limit of water solubility [30].

Response surface methodology (RSM) has been used in many studies for the optimization of many industrial processes. The RSM offer less time-consuming and do include interactive effects among the variables and response. D-optimal design is one of the reliable common RSM mode of optimization. D-optimal designs is generated for multi-factor experiments with both quantitative and qualitative factors. D-optimal designs are constructed to minimize the generalized variance of the estimated regression coefficients. In the multiple regression setting, D-optimal designs minimize the overall variance of the estimated regression coefficients [31]. As a result, the 'optimality' of a given D-optimal design is model dependent. The advantages of this model have been chosen to optimize multi-discipline works such as in building model and multidisciplinary [31], drug delivery [32], milling cutting fluid [33], separation and isolation of fatty acid [34], plant oil hydrolysis [35,36], epoxide ring opening [37], mineralization and discoloration of dye reaction [38], synthesis of plant-based synthetic esters for biolubricants [27] etc.

Table 1
Parameters and levels for D-optimal design of the synthesis parameters.

Parameter	Unit	Symbol	Variable levels		
			−1	0	+1
CPOFAs:NPG	mole	A	2	2.5	3
Time	h	B	2	4	6
Temperature	°C	C	120	140	160
H_2SO_4	% (wt)	D	1	2	3

The aim of our research was to synthesis of polyol ester-based green biolubricant derived from crude palm oil fatty acids (CPOFAs) and polyhydric alcohol of neopentyl glycol, NPG through esterification process. Response surface methodology (RSM) D-optimal design as a systematic interactions approach was used to optimize and evaluate the conversion of CPOFAs into final ester product of crude palm oil fatty acids-neopentyl glycol diester (CPOFAs-NPGDE). The effects of esterification process conditions toward the esterification responses (yield and composition of diester percentages) were optimized. The lubrication, tribology and rheology properties of the synthesized CPOFAs-NPGDE were evaluated.

2. Materials and methods

2.1. Materials

Crude palm oil (CPO) was obtained from a local refinery, Sime Darby Plantation Berhad (647766-V), located at the Carey Island, Selangor, Malaysia. Neopentyl glycol (97%) was purchased from Sigma Aldrich Chemical (Germany). Sulfuric acid, toluene, ethyl acetate, sodium bicarbonate, sodium chloride, sodium sulfate, hydrochloric acid, *n*-hexane, potassium hydroxide and ethanol were purchased from Systerm Chemicals Co (Malaysia). All chemicals were analytical reagents and used without further purification.

2.2. Synthesis of biolubricant

CPO was first hydrolyzed to produce CPOFAs according to Japir et al., 2021 [39]. The CPOFAs was esterified with neopentyl glycol, NPG to produce the NPG polyol ester biolubricant base stocks. D-Optimal design was implemented in each experiment run according to Table 1 and the primary determining factors that would influence the outputs/responses were ratio of CPOFAs to polyhydric alcohol (mol:mol), and reaction time (h), reaction temperature (°C) and catalyst amount (%) were evaluated. Each esterification process was carried in excess CPOFAs manner to ensure the NPG was completely esterified. A Dean-Stark three-necked flask (500 ml) equipped with a magnetic stirrer, a thermometer, and a reflux condenser was used for the esterification reaction. About 200 ml of toluene (azeotropic agent) was added gradually into the mixture prior to completion of the esterification. After reaction termination, the heating was turned off and the reaction product was left to cool at room temperature. The cooled reaction product was dissolved with 25 ml of ethyl acetate and transferred into a 100 ml separation funnel. 20 ml of saturated sodium bicarbonate ($NaHCO_3$) was added to the separation funnel and shaken. Then, the funnel separator was left until two layers formed. The aqueous layer at the bottom was removed, leaving the organic layer. The organic layer was further washed three times with a solution of sodium bicarbonate ($NaHCO_3$). Subsequently, the organic layer was again, washed with 10 ml of saturated sodium chloride (NaCl) and 10 ml of distilled water twice, in order to avoid any emulsion formation. Trace CPOFAs was removed by dissolving in excess ethyl alcohol. Once the two layers were formed, the bottom layer was removed. The organic layer washing was repeated until a solution with pH 7 was obtained. Then, the washed organic layer was poured into a round flask and rotary-evaporated at 90–100 °C to remove ethyl acetate and ethyl alcohol solvent giving polyol ester product in the form of a viscous liquid. The product was placed in a small beaker and sodium sulfate (Na_2SO_4) was added to absorb the remaining water in the sample. The hydrated Na_2SO_4 was filtered off prior to the product analysis.

2.3. Determination of hydroxyl value

The amount (mg) of KOH required to neutralize acetic acid released from the acetylation of the hydroxyl group present in 1 g of oil or fat is indicated by the hydroxyl value (or acetyl value). Therefore, the number of OH groups in fat or oil indicates the hydroxyl value. Variation in hydroxyl content can provide an indication of CPO quality, where a high content of hydroxyl indicates poor oil quality. The increased presence of free hydroxyl groups in crude palm kernel oil can be attributed to the breakdown of TAG by the action of microbial lipase and liberated FFA, DAG, and MAG. The hydroxyl value was calculated using eq. (1) as according to AOCS Cd 13–60 method [40].

$$\text{Hydroxylvalue} = \frac{(V_b - V_s) \times N \times 56.1}{W} + \text{AV} \quad (1)$$

Where,

V_b is the volume (mL) of the KOH solution used for the blank

V_s is the volume (mL) of the KOH solution used for the determination of the sample

N is the normality of the KOH (Eq/L).

AV is the acid value of the sample, 56.1 is the molecular weight of KOH

W is the weight (g) of the sample test portion

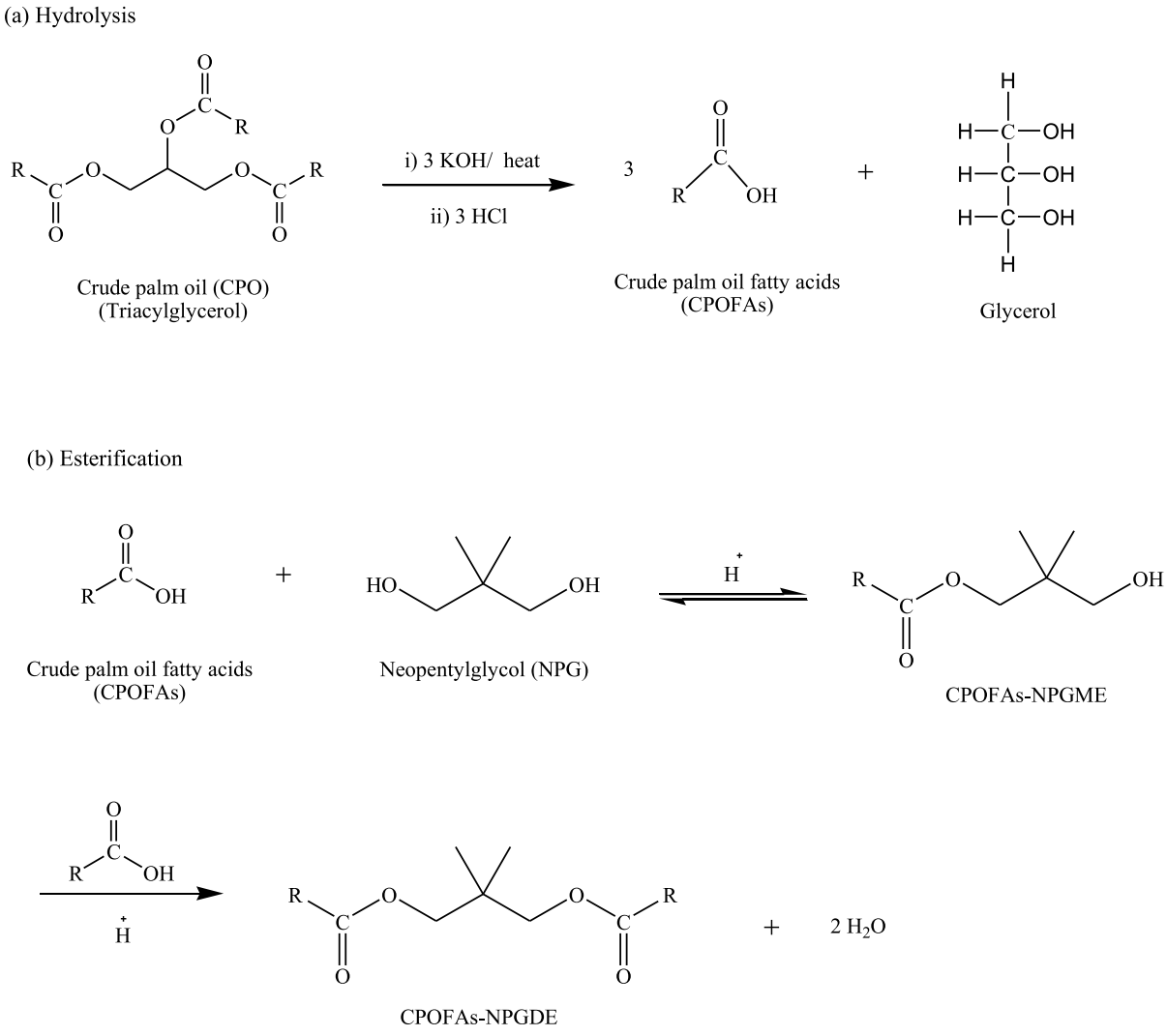
2.4. Esterification optimization

There were four factors have been identified as independent parameters that affected the esterification process namely moles of CPOFAs to NPG, reaction time, reaction temperature and the amount of H_2SO_4 catalyst as shown in Table 1. The values of the variables were selected based on conventional factor-by-factor optimization screening tests (unpublished data). The screening tests showed that high percentages of polyol esters yield (65–85) were observed at NPG:POFAs molar ratio between 1:2–3.5, 1–4% catalyst, reaction temperature of 120–180 °C and reaction time of 2–6 h. These values were further employed and varied within the lowest and highest values in the D-optimal design optimization for the highest values of the reaction responses, polyol diester yield percentages (Y_1) and selectivity (Y_2).

The implementation of the esterification experiments based on the D-optimal design has generated 25 runs for the synthesis of crude palm oil fatty acids-based neopentyl glycol diesters biolubricant (CPOFAs-NPGDE). Each individual experimental was carried out according to the reaction variable parameters suggested by the model design to obtain it reaction responses of yield and product selectivity percentages.

2.5. Polyol ester molecular characterization

The structure of the diester was confirmed and determined using Nuclear Magnetic Resonance (NMR) and Fourier Transform infrared spectroscopy (FTIR). ^1H and ^{13}C NMR images were recorded on JEOL-ECP 400 spectrometer (400 MHz ^1H/100.61 MHz^{13}C) using $CDCl_3$ as the solvent. The FTIR spectra were recorded on a PerkinElmer Infrared Spectrophotometer within the range between 500 and 4000 cm^{-1}. The composition of fatty acids was determined by using GC-FID (Shimadzu, Series GC-17A) equipped with BPX 70 polarized capillary column (30m × 0.25 mm × 0.25 μm thinning thickness; SGE) and FID detector. Fatty acid methyl ester (FAME) was prepared from acid-catalyzed transesterification according to Jumaah et al. [27]. The parameters of GC were carried out according to Nor et al. [35]. The injector and detector temperature were set at 250 and 280 °C respectively. The column



R = Myristic acid $C_{14:0}$, Palmitic acid $C_{16:0}$, Stearic acid $C_{18:0}$, Oleic acid $C_{18:1}$ and Linoleic acid $C_{18:2}$

Fig. 1. Schematics representation for (a) CPO hydrolysis and (b) esterification reaction of CPOFAs with NGP to form CPOFAs-NPGDE.

temperature was maintained at 120 °C for 1 min and then increased to 250 °C at a rate of 3 °C/min and maintained at the final temperature for 15 min. Helium gas was used as gas carrier at a flow rate of 0.3 mL/min with a total flow rate of 13 mL/min. The sample in the solvent was injected using a split injector mode with a separation ratio of 29: 1. The fatty acid composition was identified based on the retention time of Merck authentic standards FAME solution analyzed under the same condition.

2.6. Lubrication appearances

The standard methods of the American Society for Testing Materials (ASTM) were used to determine the lubrication characteristics. The ASTM D-6186 of oxidative stability [41], ASTM D-97 of pour point [42], ASTM D-445 of kinematic viscosities [43], ASTM D-92 of flash point [44], ASTM D-2270 of viscosity index [45] and ASTM D 6168 of oxidative stability (oxidation onset temperature) [41] were used. The analyses were carried out in triplicates, and the results are the averages of the three values. The ASTM method D4172-94 [46] was used to assess the tribological properties of the products. Rheological properties (Newtonian or Non-Newtonian) are determined using Anton Paar Physica MCR 301 model (Germany) rheometer with cone and plate geometry [47,48]. A 0.051 mm diameter CP 25–2 cone spindle was used. A shear rate ranging from (0–100) s^{-1} was used at a temperature of 25 °C. The effect of shear rate (γ) on shear stress (τ) is studied and the type of fluid is determined based on the resulting shear stress trend. Anton Paar Physica MCR 301 model (Germany) rheometer with a four-ball geometry was used for tribological properties. A sliding speed between 10^{-1}–10^{3} mm/s was deployed at 40 °C and 100 °C. Friction coefficient of biolubricant tribological properties were performed according to ASTM method D4172-94 and discussed as their Stribeck curves at 40 °C and 100 °C [49]. The Stribeck curve was used to define the tribological properties of the biolubricant.

3. Results and discussion

3.1. CPOFAs-NPG diester synthesis

CPO was first alkaline hydrolyzed to produce crude palm oil fatty acids (CPOFAs) according to Nor et al. [35]. The CPO hydrolysis process involved two stages of CPO conversion to soap and followed by the acidification processes to produce CPOFAs. About 95% wt. of CPOFAs

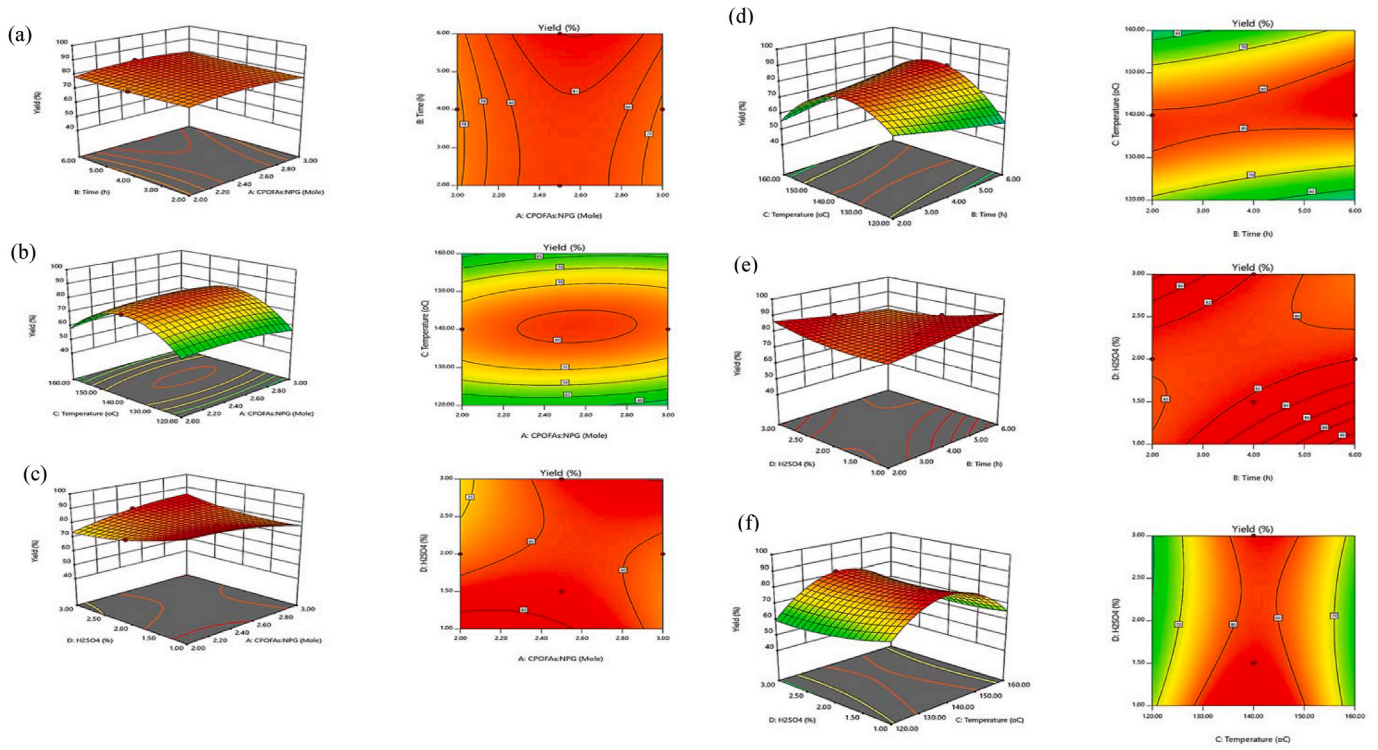


Fig. 2. 3-D surface and contour plots of CPOFAs-NPGDE yield (%) at different parameters interaction: (a) CPOFAs:NPG molar ratio vs reaction time; (b) CPOFAs:NPG molar ratio vs reaction temperature; (c) CPOFAs:NPG molar ratio vs catalyst amount; (d) reaction time vs reaction temperature; (e) reaction time vs catalyst amount; (f) reaction temperature vs catalyst amount.

Table 2
Experimental runs from D-optimal design and the respective responses of the CPOFAs-NPGDE.

Run	Factors				Responses	
	CPOFAs: NPG (mole: mole)	Reaction time (h)	Reaction temp. (°C)	Catalyst amount (%)	Yield, Y_1 (%)	Selectivity, Y_2 (%)
1	2.00	6.00	120.00	1.00	71	87
2	3.00	2.00	120.00	3.00	73	93
3	2.00	6.00	120.00	3.00	41	59
4	2.50	2.00	140.00	2.00	80	96
5	3.00	6.00	160.00	3.00	77	97
6	2.00	6.00	160.00	2.00	67	98
7	3.00	6.00	120.00	2.00	51	68
8	2.00	4.00	140.00	2.00	78	99
9	2.00	2.00	160.00	1.00	53	92
10	2.00	2.00	160.00	1.00	53	92
11	2.50	4.00	140.00	1.50	83	100
12	3.00	2.00	160.00	2.00	55	80
13	3.00	6.00	160.00	1.00	75	93
14	2.50	4.00	140.00	3.00	82	98
15	2.00	2.00	160.00	3.00	54	80
16	3.00	4.00	120.00	1.00	59	79
17	3.00	2.00	140.00	1.00	73	97
18	2.50	6.00	140.00	2.00	82	95
19	2.50	2.00	120.00	1.00	71	88
20	3.00	6.00	160.00	1.00	74	92
21	2.00	2.00	120.00	2.00	68	78
22	3.00	2.00	120.00	3.00	73	91
23	3.00	6.00	160.00	3.00	77	95
24	2.00	6.00	120.00	1.00	70	86
25	3.00	4.00	140.00	2.00	78	97

were produced. The resulting CPOFAs was then esterified with polyhydric alcohol of NPG to produce crude palm oil fatty acids-NPG diester (CPOFAs-NPGDE). The esterification reaction of CPOFAs with NPG was first produced monoester (CPOFAs-NPGME) as intermediate and further converted to diester (CPOFAs-NPGDE) as shown in Fig. 1.

3.2. Optimization process

The polyol diester CPOFAs-NPGDE was synthesized via acid-catalyzed esterification process of CPOFAs with NPG. The esterification process was optimized by using response surface methodology of D-optimal design to obtain the highest percentages of the reaction responses of the CPOFAs-NPGDE yield and selectivity. The experimental design was established in accordance to three levels factorial and four reaction factors which involved the molar ratio of CPOFAs:NPG (A), reaction time (B), reaction temperature (C) and catalyst amount (D), respectively. Each parameter with their levels, uncoded experimental design and their process responses from the software are tabulated in Tables 1 and 2. The esterification reaction responses (percent yield and selectivity) of the CPOFAs-NPGDE product were statistically analyzed and the results were represented by varying plots showing the main and important interaction effects between the reaction variables parameters. 3D surface view plots and their contour plots were integrated based on the four independent factors. Moreover, empirical models and mathematical models were calculated and optimized by using D-optimal design RSM software.

3.3. Evaluation of the biolubricant synthesis

The percentage yield and the selectivity of CPOFAs-NPGDE product obtained from the D-optimal design were evaluated for each individual variable parameters suggested. The main and interaction effect plots concerning all esterification parameters involving CPOFAs:NPG molar ratio, reaction time, reaction temperature and catalyst are discussed. Table 2 represents tabulated 25 experimental run based on the suggested design data at selected reaction parameters from the D-optimal design matrix and their experimental respective reaction responses of CPOFAs-

NPGDE yield (Y_1) and selectivity (Y_2). The yield of CPOFAs-NPGDE were ranged from 41% the lowest up to the highest of 83%. The D-optimal model analysis showed the yield (Y_1) has good response with the reaction factors and the power transform has little effect with ratio max to min value of 2.0244. Whereas the selectivity of the CPOFAs-NPGDE showed high percentages from 59% up to 100%. The model analysis showed the selectivity (Y_2) has good response with the reaction factors and the power transform has little effect with ratio max to min value of 1.6949, respectively. Therefore, the RSM model design suggested for the highest order of the quadratic polynomial model with significant effect and the model is not aliased or misidentify.

The plot for the residual versus experimental run number can be seen in Supplementary materials Fig. S1. There is no evidence to show that the process was being drifted. It rather shows reliable results as the outcome of a competent experimental conduct. The fact that the plots scattered within the specified domains and not exceeded indicate that the model would be accepted for validation. The residuals are randomly positioned on the upper and downside of the straight centre line, approximately between −2.00 and 2.00 for reaction responses. Therefore, the experimental run errors are normally distributed.

The agreement between the actual experimental data against the predicted values from D-optimal design model for the yield and selectivity of CPOFAs-NPGDE, respectively (Supplementary materials Fig. S2). It is observed that all data points clustered around the line, suggested that the empirical data and the models were indeed comparable. As it is, the models able to forecast the predicted values of CPOFAs-NPGDE yield and CPOFAs-NPGDE selectivity which remarkably coincide with the actual experimental values. The model fit with sequential p-value of <0.0001 and lack of fit of 0.3191 are reliable for the prediction of the CPOFAs-NPGDE yield. Furthermore, the same model fit with sequential p-value of <0.0001 and lack of fit of 0.4710 are consistent for the prediction of the CPOFAs-NPGDE selectivity. Statistically, the values of R^2 (0.9992) and Adjusted R^2 (0.9981) for the yield and R^2 (0.9969) and Adjusted R^2 (0.9926) for the diester selectivity, suggest the precision of the model is certainly dependable. Hence, the idea of proceeding with the next stage of analysis via the optimization tool is desirable.

3.4. Model design fitting

The ANOVA data analysis for the yield and the selectivity of CPOFAs-NPGDE are shown in Table 3. The CPOFAs-NPGDE yield fits the model F-value of 894.64 that implies the model is significant. There is only a 0.01% chance that an F-value this large could occur due to noise. The CPOFAs-NPGDE selectivity shows model F-value of 230.49 implies the model is also significant. It means only a 0.01% chance that an F-value this large could occur due to noise. The statistical result also showed that the difference between predicted R-squared, R^2 (0.9904) and Adjusted R^2 (0.9981) of the CPOFAs-NPGDE yield is in reasonable agreement. The detail statistical analysis can be seen in Supplementary Material of Table S1. Furthermore, the difference between predicted R-squared, R^2 (0.9803) and Adjusted R^2 (0.9926) of the CPOFAs-NPGDE selectivity is also in good agreement with less than 0.2. The model shows the Adequate Precision ratio (greater than 4 is desirable) of 105.33 for the yield of CPOFAs-NPGDE and of 60.048 for the selectivity of CPOFAs-NPGDE indicates adequate signal measures the signal to noise ratio. These indicated the models are satisfactory. Moreover, relative to the pure error, the lack of fit of the models (CPOFAs-NPGDE yield and CPOFAs-NPGDE selectivity) showed F-values of 1.56 and 0.54, respectively and are considered insignificant on the basis of their P-values ($P > 0.05$). These indicate both models are satisfactory fitting. The ANOVA statistical analysis can be seen in Supplementary Material of Table S2. The regression line equations suggested by a quadratic model design for CPOFAs-NPGDE yield (Y_1) and CPOFAs-NPGDE selectivity (Y_2) are shown in Equations 2 and 3, respectively.

Table 3
ANOVA analysis data and regression model for synthesis of CPOFAs-NPGDE.

Responses	ANOVA value	Model	Factors and their interaction													Lack of fit
			A	B	C	D	AB	AC	AD	BC	BD	CD	A^2	B^2	C^2	D^2	
CPOFAs-NPGDE Yield	Sum of Squares	3204.48	4.15	8.35	17.77	48.38	1.57	47.97	342.02	763.65	295.56	82.71	14.65	0.4512	571.67	37.77	1.56
	df	14	1	1	1	1	1	1	1	1	1	1	1	1	1	1	5
	Mean Square	228.89	4.15	8.35	17.77	48.38	1.57	47.97	342.02	763.65	295.56	82.71	14.65	0.4512	571.67	37.77	0.3117
	F-Value	894.64	16.22	32.63	69.44	189.09	6.12	187.51	1336.8	2984.7	1155.2	323.28	57.26	1.67	2234.4	147.62	1.56
	P-value	0.0001	0.0024	0.0002	0.0001	0.0001	0.0329	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.2137	0.0001	0.0001	0.3191
	Status	***	***	***	***	***	***	***	***	***	***	***	***		***	***	
	$R^2 = 0.9992$, Adjusted $R^2 = 0.9981$, Predicted $R^2 = 0.9904$, adequate precision = 105.326																
	Regression model	CPOFAs-NPGDE Yield (Y_1) = 80.7932 + 0.523652A + 0.743924B + 1.19124C - 1.81348D + 0.364547AB + 1.97625AC + 5.29772AD + 7.86602BC - 4.9192BD + 2.65166CD - 2.56276A^2 - 18.5668C^2 + 2.96538D^2															… (4)
CPOFAs-NPGDE Selectivity	Sum of Squares	2488.29	6.61	0.0584	515.75	1.8607	30.65	38.57	491.44	534.00	69.49	23.85	1.21	2.63	410.44	82.89	2.71
	df	14	1	1	1	1	1	1	1	1	1	1	1	1	1	1	5
	Mean Square	177.73	6.61	0.0584	515.75	1.8607	30.65	38.57	491.44	534.00	69.49	23.85	1.21	2.63	410.44	82.89	0.5422
	F-Value	230.49	8.57	0.0757	668.83	241.31	39.75	50.01	637.31	692.50	90.11	30.93	1.57	3.41	532.27	107.49	0.5422
	P-value	0.0001	0.0151	0.7888	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002	0.2382	0.0944	0.0001	0.0001	0.7410
	Status	***	***		***	***	***	***	***	***	***	***			***	***	
	$R^2 = 0.9969$, Adjusted $R^2 = 0.9926$, Predicted $R^2 = 0.9803$, adequate precision = 60.048																
	Regression model	CPOFAs-NPGDE Selectivity (Y_2) = 97.0181–0.660611A + 6.41837C - 3.55655D −1.61299AB - 1.77192AC + 6.35034AD + 6.57775BC - 2.38517BD + 1.42394CD - 15.7323C^2 + 4.39304D^2															… (5)

Note: A = CPOFAs:NPG molar ratio; B = Reaction time; C = Reaction temperature; D = Catalyst amount; *** = Significant at $P < 0.05\%$ level.

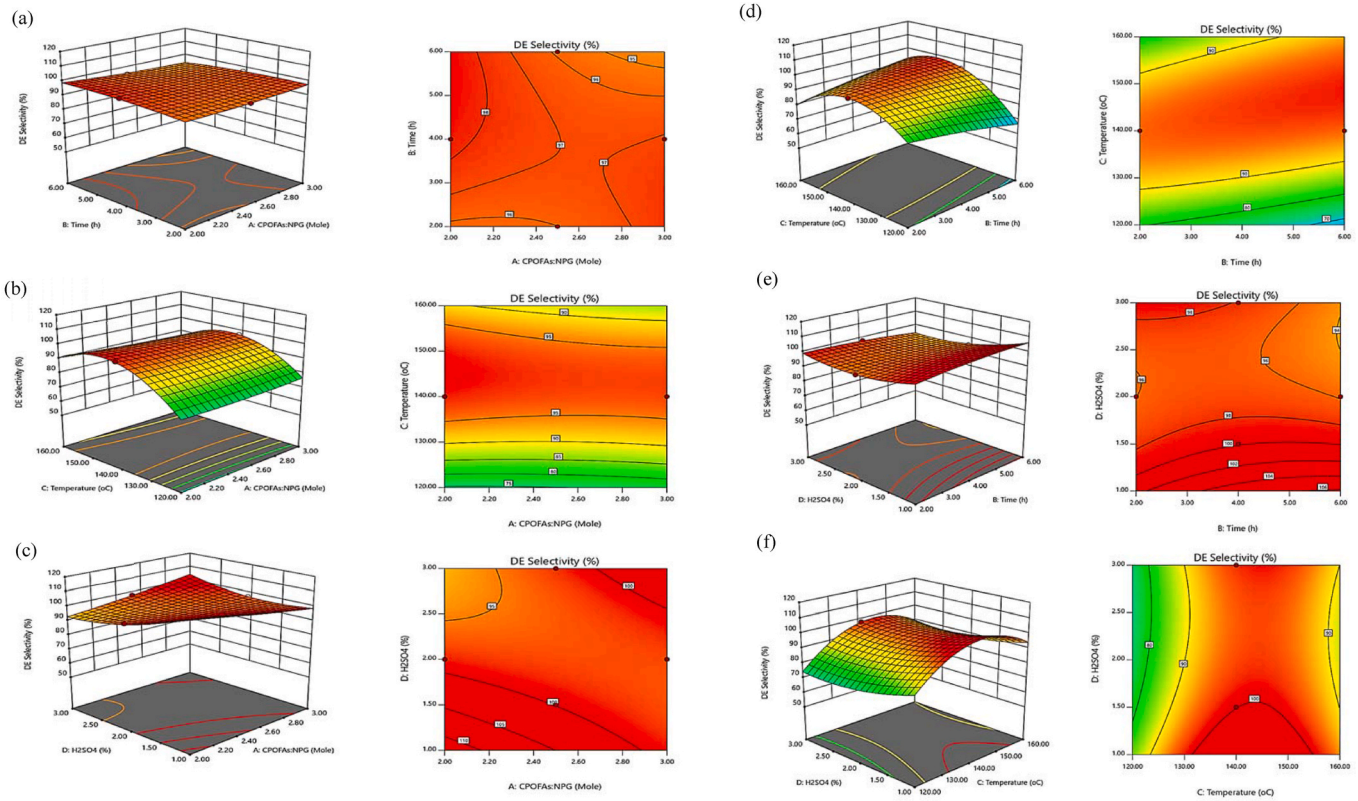


Fig. 3. 3-D surface plots and contour plots for selectivity of CPOFAs-NPGDE (%) at different parameters interaction: (a) CPOFAs:NPG molar ratio vs reaction time; (b) CPOFAs:NPG molar ratio vs reaction temperature; (c) CPOFAs:NPG molar ratio vs catalyst amount; (d) reaction time vs reaction temperature; (e) reaction time vs catalyst amount; (f) reaction temperature vs catalyst amount.

$$\text{CPOFAs-NPGDE Yield } (Y_1) = 80.7932 + 0.523652A + 0.743924B + 1.19124C - 1.81348D + 0.364547AB + 1.97625AC + 5.29772AD + 7.86602BC - 4.9192BD + 2.65166CD - 2.56276A^2 + 0.452841B^2 - 18.5668C^2 + 2.96538D^2 \ldots. \ldots. \ldots. \quad (2)$$

$$\text{CPOFAs-NPGDE Selectivity } (Y_2) = 97.0181–0.660611A - 0.0622133B + 6.41837C - 3.55655D - 1.61299AB - 1.77192AC + 6.35034AD + 6.57775BC - 2.38517BD + 1.42394CD + 0.737625A^2 - 1.09365B^2 - 15.7323C^2 + 4.39304D^2 \ldots. \ldots. \ldots. \quad (3)$$

ANOVA analysis for the yield of CPOFAs-NPGDE in Table 3 indicated that the model terms for parameters of CPOFAs:NPG molar ratio (A), reaction time (B), reaction temperature (C) and catalyst amount (D) factors are significant based on their P-values ($P < 0.05$). The interaction effect between the parameters of AB, AC, AD, BC, BD and CD are measured as significant ($P < 0.05$). Moreover, the mean square values of the parameters A^2, C^2 and D^2 are also significant within $P < 0.05$ except B^2. Therefore, the final regression line equations for CPOFAs-NPGDE yield (Y_1) for a polynomial quadratic model is shown in Equations 4. Furthermore, the ANOVA analysis for the CPOFAs-NPGDE selectivity indicated that the interaction effect between the parameters of A, C and D (except B), and their respective square values of C^2 and D^2 (except A^2 and B^2) are significant factors within their P value < 0.05. Likewise, the interaction between AB, AC, AD, BC, BD and CD are restrained as significant on the CPOFAs-NPGDE selectivity based on the P value < 0.05. Therefore, the final regression line equation for prediction of the CPOFAs-NPGDE selectivity (Y_2) of the quadratic model is shown in Equation 5.

The core effects toward the response yield (Y_1) and selectivity of CPOFAs-NPGDE (Y_2) shown by CPOFAs:NPG molar ratio (A), reaction time (B), reaction temperature (C) and catalyst amount (D) whereas no interaction effect was plotted (Supplementary materials Figure S3). The relationship effect of the input parameters under the esterification reaction which is connected by a line or a curve on the reaction response Y_1 and Y_2. The perturbation plots provide useful demonstration on the connection of CPOFAs:NPG molar ratio (A), reaction time (B), reaction temperature (C) and catalyst amount (D) on both the response CPOFAs-NPGDE yield (Y_1) and selectivity (Y_2), and to compare the significant effects of all the factors at a particular point in the design space. The main significant effect of the parameters is portrayed by a line with a steeper slope for long-range change as compared with by less significant effects contributed factors. The plot for a yield of CPOFAs-NPGDE ester (Y_1) against molar ratio of CPOFAs:NPG (A) and reaction time (B) shows a positively significant effect from a low level to a high level (plot (a)). However, the plot for Y_1 against catalyst amount (D) shows a significant negative effect from a low level to a high level and reaction temperature (C) shows a significant negative effect from a low level to a middle level, respectively. However, the plots of molar ratio of CPOFAs:NPG (A) and reaction time (B) have a lower slope than reaction temperature (C) and catalyst amount (D). This plot indicated that the reaction temperature (C) and catalyst amount (D) factors contribute more significant effect among the other parameters of CPOFAs:NPG molar ratio (A) and reaction time (B). Moreover, the plot illustrates a steeper curvature for reaction temperature (C) toward the yield of the ester. This curvature slope of the yield of the ester versus reaction temperature (C) showed that the yield of the ester value (Y_1) is sensitive and substantially increases when reaction temperature increased from 120 °C to 140 °C. This result showed that the value of yield of ester significantly increased from 59% to 83% when reaction temperature increased from low level of 120–140 °C within the low level of catalyst amount. The main effect plot for the response of CPOFAs-NPGDE selectivity (Y_2) towards the reaction parameters which shows the most influence by reaction temperature (C) followed by catalyst amount (D), reaction time (B) and CPOFAs:NPG molar ratio (A) (plot (b)). As shown by the response of the ester yield, a

steeper curvature in a reaction temperature (C) shows that the selectivity of CPOFAs-NPGDE (Y_2) is more sensitive towards the reaction temperature (C). This curvature slope of the selectivity of CPOFAs-NPGDE versus reaction temperature showed that the selectivity of CPOFAs-NPGDE value substantially increased from 68% to 100% when reaction temperature increased from low level of 120–140 °C within the low level of catalyst amount. Therefore, the reaction temperature (C) is the most significant factor that influences the value of both the yield and the selectivity of CPOFAs-NPGDE ester product.

3.5. Response surface analysis

The collaboration effect between the reaction variables on the CPOFAs-NPGDE diester yield (Y_1) can be demonstrated by using 3-D response surfaces and contour graphs. Fig. 2(a-c) shows the 3-D and contour plots for the interaction between CPOFAs:NPG molar ratio and reaction time (AB), CPOFAs:NPG molar ratio and reaction temperature (AC), and CPOFAs:NPG molar ratio and catalyst amount (AD), respectively. The 3-D surface contour has illustrated the maximum yield of CPOFAs-NPGDE ester (Y_1) can be reached in a middle region molar ratio of 2.4 CPOFAs:NPG, at a middle level of 4 h reaction time, at a lower level of 1% catalyst amount, and at the middle of 140 °C reaction temperature. Additionally, the 3-D response surface contour plots for the interaction effect between reaction time and reaction temperature (BC), reaction time and catalyst amount (BD), and reaction temperature and catalyst amount (CD) on the yield of CPOFAs-NPGDE ester (Y_1) is shown in Fig. 2(d-f), respectively. It is noteworthy that the same trend where the highest yield of CPOFAs-NPGDE ester can be facilitated at middle reaction time of 4 h, middle reaction temperature of 140 °C and in the low region of 1.0% catalyst amount. Thus, the desired optimal region to reach maximum yield of CPOFAs-NPGDE ester (Y_1) at 83% is at 4 h reaction time, and reaction temperature of 140 °C, reactants molar ratio of 2.4 with a 1% of catalyst amount. As discussed earlier, the yield (Y_1) was the most sensitive and effected towards the reaction temperature (C) as compared to other parameters. It is noteworthy that when the reaction temperature was increased from 120 to 140 °C, the CPOFAs-NPGDE yield start to increase from 65% to 83%. Nevertheless, when the reaction temperature reached at 160 °C, the CPOFAs-NPGDE yield reduced back to about 67%. This may be attributed to the diester breaking down due to persistently occurring reverse reactions to a lower monoester conversion. This finding is in agreement with Mamud et al. [36] that confirmed the triesters of trimethylopropane triolates (TMPTE) yield was increased with the reaction temperature. However, when the temperature increases to higher temperature of 180 °C, the triester formation was dropped.

Fig. 3 displays the 3-D plots and contour plots for the relation and interaction between the input parameter factors and the selectivity of CPOFAs-NPGDE ester (Y_2). Fig. 3(a–c) shows the interaction between the molar ratio of CPOFAs:NPG and reaction time (AB), molar ratio of CPOFAs:NPG and reaction temperature (AC), molar ratio of CPOFAs:NPG and catalyst amount (AD). Generally, it is clear that the maximum selectivity of CPOFAs-NPGDE ester (Y_2) can be achieved when the catalyst amount in it lower level (1%) and other parameters (molar ratio of CPOFAs:NPG, reaction time and reaction temperature) in their middle region of 2.40 mol, 140 °C and 4h, respectively. Fig. 3(d–f) shows the 3D and contour plots for the interaction effect between reaction time and reaction temperature (BC), reaction time and catalyst amount (BD), and reaction temperature and catalyst amount (CD) on the selectivity for CPOFAs-NPGDE (Y_2). It is perceived that the highest selectivity for CPOFAs-NPGDE can be accomplished at low value of catalyst amount (1%) and the mean value of reaction temperature (140 °C) and reaction time (4 h). Therefore, the desired optimal region to reach maximum selectivity of CPOFAs-NPGDE ester (Y_2) at 100% is at 4 h reaction time, and reaction temperature of 140 °C, reactants molar ratio of 2.40 with a 1% of catalyst amount. At this optimal condition diester breakdown via persistently occurring reverse reactions to a lower monoester (CPOFAs-NPGME ester) at high temperature and excess amount of catalyst is reduced.

Table 4
Restraints for optimization of CPOFAs-NPGDE synthesis parameters and responses for RSM.

Parameters and Response	Goal	Lower limit	Upper limit	Weight	Importance
A: Ratio of CPOFAs:NPG (mol:mol)	is in range	2	3	1	3
B: H_2SO_4 (%)	is in range	1	3	1	3
C: Reaction temp (°C)	is in range	120	160	1	3
D: Reaction time (h)	is in range	2	6	1	3
Yield of CPOFAs-NPGDE	Maximize	45 ± 3	83 ± 3	1	3
Selectivity of CPOFAs-NPGDE	Maximize	72 ± 2	100 ± 3	1	3

3.6. D-Optimal design model validation

A numerical optimization was performed to seek the optimal condition for synthesizing the CPOFAs-NPGDE as polyol ester-based biolubricant. All controlling parameters of A, B, C and D in a set of ranges were selected to achieve maximum yield (Y_1) and selectivity (Y_2), where the higher the desirability of the functions, the better the model's accuracy. Table 4 summarizes all of the limitations applied to find the optimal settings for the CPOFAs-NPGDE yield and selectivity. The optimal setting for input parameters was set in a range value whereas the optimal setting for the CPOFAs-NPGDE yield and selectivity was set at the maximize value. The estimated predicted factors from the model numerical optimization for synthesizing the CPOFAs-NPGDE is shown in Fig. 4. The optimum condition forecast through the D-optimal model design generated for CPOFAs:NPG molar ratio was at 2.26:1 (mol:mol), reaction time of 4.79 h, reaction temperature of 138.55 °C and H_2SO_4 catalyst amount of 1.12%, respectively. Prearranged these conditions, the yield of CPOFAs-NPGDE was predicted at 87.93%, and the selectivity of CPOFAs-NPGDE was at 100%.

Table 5 represents five replicated experiments done for verification and to ensure the accuracy of the predicted model values at the optimal reaction conditions. These experiments produced an average of 87.6 ± 0.5% yield of the CPOFAs-NPGDE and 100.0 ± 0.5% selectivity of the CPOFAs-NPGDE, respectively. These confirmation results appeared to be in good agreement with the predicted values of 87.9% yield and 100% selectivity of CPOFAs-NPGDE. Therefore, this study confirms that the D-optimal design is a reliable, simple, and useful approach for assessing the best conditions to synthesize the CPOFAs-NPGDE. This study showed that response surface methodology (RSM) of D-optimal design produced higher product yield as compared the yield without optimization (83%) as shown in experimental run number 11 (Table 2). The study yield result is in agreement with 87.7% NPG diester from high oleic palm oil methyl ester (HOPME) [23]. However, the later used higher esterification temperature of 182 °C as compared to lower temperature of 138 °C for this study. Moreover, the yield result is also in comparable and acceptable due to its low cost operational as compared to 95% NPG dieaters from castor and soybean biodiesel through enzymatic process with longer esterification time of 24 h [26] and 99% NPG diester from hexanoic acid with higher temperature of 217.2 °C [28].

3.7. Polyol ester composition

In the synthesis of CPOFAs-NPGDE, monoester (CPOFAs-NPGME) was first formed as an intermediate before further esterified to form diester of CPOFAs-NPGDE as illustrated in Fig. 1. The composition

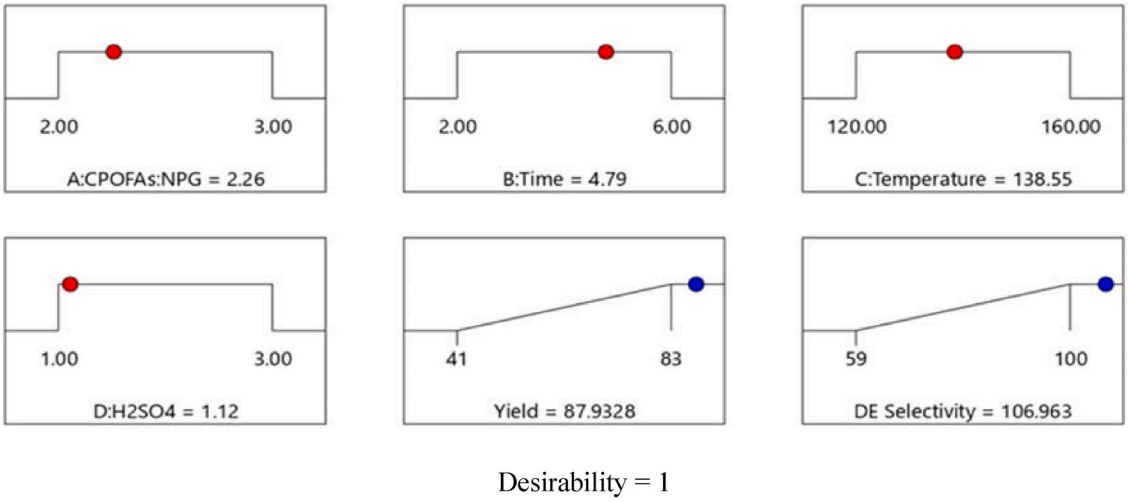


Fig. 4. RSM predicted model at optimal conditions for the synthesis of CPOFAs-NPGDE.

Table 5
Validation test result at the optimum conditions for CPOFAs.

	Independent variables					Responses	
	No	A	B	C	D	Y_1	Y_2
Actual	1	2.26:1	4.8	138.0	1.12	87.5 ± 1.0	100.0 ± 1.3
	2	2.26:1	4.8	138.0	1.12	87.9 ± 1.3	100.0 ± 1.0
	3	2.26:1	4.8	138.0	1.12	87.4 ± 1.0	100.0 ± 1.2
	4	2.26:1	4.8	138.0	1.12	87.8 ± 1.1	100.0 ± 1.2
	5	2.26:1	4.8	138.0	1.12	87.6 ± 1.1	100.0 ± 1.3
Actual average		2.26:1	4.8	138.0	1.12	87.6 ± 1.1	100.0 ± 1.2
Predicted		2.26:1	4.8	138.0	1.12	87.9	100.0

Note: A = CPOFAs:NPG molar ratio (mol:mol); B = Reaction time (h); C = Reaction temperature (°C); D = Catalyst amount (H_2SO_4) (%); Y_1 = Yield of CPOFAs-NPGDE (%); Y_2 = Selectivity of CPOFAs-NPGDE (%).

percentage of the final polyol diester or the selectivity of CPOFAs-NPGDE was determined by high-temperature column (DB-5HT) GC-FID analysis. Fig. 5 shows the DB-5HT column GC chromatogram for CPOFAs-NPGDE. Each peak in the GC chromatogram was represented by diacyl group carbon numbers that attached to the NPG back bone molecule.

In this study, the esterification of CPOFAs and NPG was successfully produced 100% diester. This indicated that the entire hydroxyl group (OH) in NPG was successfully esterified with CPOFAs. The result was supported by the low hydroxyl value of 24.68 mgKOH/g for CPOFAs-NPGDE as compared to NPG hydroxyl value of 1075 mgKOH/g. The hydroxyl value of the synthesized CPOFAs-NPGDE was also insignificantly different from CPOFAs hydroxyl value of 19 mgKOH/g. This indicated that the esterification was 100% successfully converted NPG into CPOFAs-NPGDE with respect toward their hydroxyl value.

GC-FID analysis showed that CPOFAs-NPGDE consists of four types of diesters namely 1.34% diester 30 (DE30), 26.2% diester 32 (DE32), 48.46% diester 34 (DE34) and 24% diester 36 (DE36). DE30 is a diester in which the acyl chain group consists of myristate (C_{14}) and palmitate (C_{16}). DE32 composes of both palmitate (C_{16}) acyl chains. DE34 consists of palmitate (C_{16}) and 18 carbon atoms acyl group either stearate ($C_{18:0}$), oleate ($C_{18:1}$) or linoleate ($C_{18:2}$). While DE36 is a diester consists of both 18 carbon atoms acyl group chains. The percentage of fatty acid

Table 6
Fatty acids composition of CPO, CPOFAs and CPOFAs-NPGDE.

Fatty acid	CPO (%)	CPOFAs (%)	CPOFAs-NPGDE (%)
Myristic, $C_{14:0}$	0.49 ± 0.10	0.47±±0.21	1.00 ± 0.21
Palmitic, $C_{16:0}$	41.78 ± 0.51	40.10 ± 0.62	46.98 ± 0.52
Stearic, $C_{18:0}$	3.65 ± 0.22	3.84 ± 0.22	6.24 ± 0.23
Oleic, $C_{18:1}$	44.41 ± 0.53	45.59 ± 0.53	41.38 ± 0.71
Linoleic, $C_{18:2}$	9.67 ± 0.21	10.00 ± 0.20	4.40 ± 0.20
Σ Saturated	45.92	44.41	54.22
Σ Unsaturated	54.08	55.59	45.78

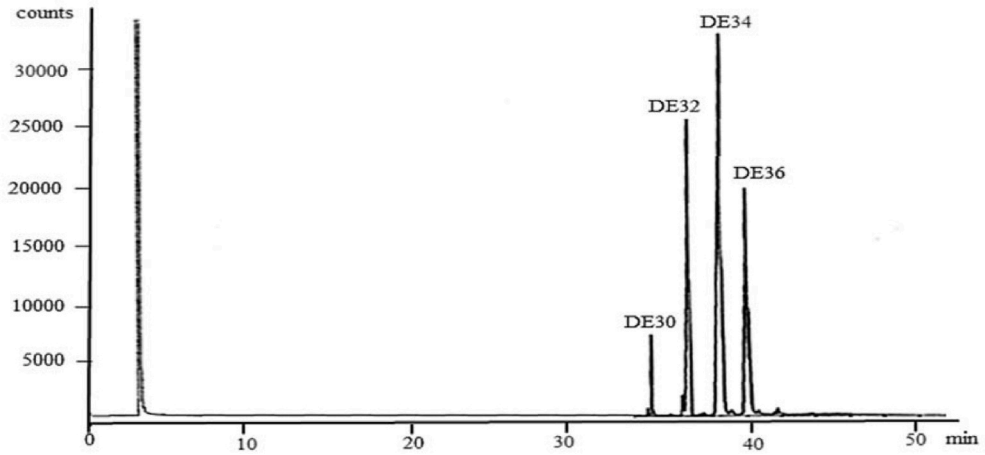


Fig. 5. High temperature column GC chromatogram for CPOFAs-NPGDE.

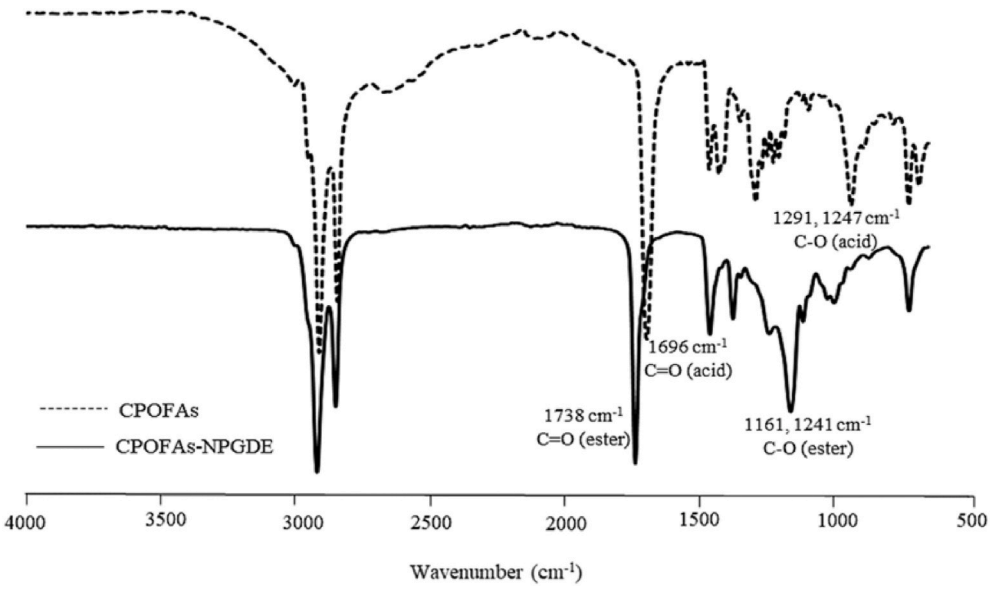


Fig. 6. FTIR spectra of CPOFAs and CPOFAs-NPGDE.

composition in CPOFAs-NPGDE was determined by GC-FID analysis using BPX-70 column. The chromatogram for CPOFAs-NPGDE fatty acid methyl ester (FAME) (Supplementary materials Fig. S4) while Table 6 shows a comparison between the fatty acid composition in CPO, CPOFAs and CPOFAs-NPGDE.

Fatty acids composition showed that CPOFAs-NPGDE composed nearly the same percentages as fatty acids composition of CPO and CPOFAs. The differences in the fatty acid composition percentage between CPO and CPOFAs are not significant. However, the composition of unsaturated acids in CPOFAs is exceeded as compared to their composition in CPO. It seems to indicate that the unsaturated fatty acids are more venerable to the enzymatic hydrolysis and presence as free fatty acids (FFA) in CPO. In general, the results show that CPO and CPOFAs contain five types of fatty acids which oleic acid as major fatty acid followed by palmitic acid, linoleic acid, stearic acid and myristic acid, respectively. CPOFAs composed the highest percentage of oleic acid at 45.59 ± 0.53%, followed by palmitic acid (40.10 ± 0.62%), linoleic acid (10.00 ± 0.20%), stearic acid (3.84 ± 0.22%) and myristic acid (0.47 ± 0.21%). The percentage ratio of total unsaturated to total saturated fatty acids in CPOFAs was 1:1 with 55.59% of unsaturated fatty acids and 44.41% of saturated fatty acids, respectively. However, the fatty acid composition in CPOFAs-NPGDE was significantly different from the raw material especially with CPOFAs. The composition of palmitic acids is dominated (46.98 ± 0.52%) and linoleic is less dominated (4.40 ± 0.20%) in CPOFAs-NPGDE. The results showed that CPOFAs-NPGDE consisted of 54.22% saturated fatty acids and 45.78% unsaturated fatty acids, which palmitic acid recorded the highest percentage of 46.98 ± 0.52%. Due to a symmetrical molecular structure of NPG, it's tendency to react with saturated fatty acids as compared to unsaturated fatty acids in CPOFAs. Straight carbon chains of saturated fatty acids have higher tendency to react with NPG due to less steric hindrance as compared to bent structure of unsaturated fatty acids with higher steric interference.

3.8. Polyol ester structural characterization

The FTIR spectra of CPOFAs (fatty acids) as the raw material and product CPOFAs-NPGDE (polyol ester) is shown in Fig. 6. The acid

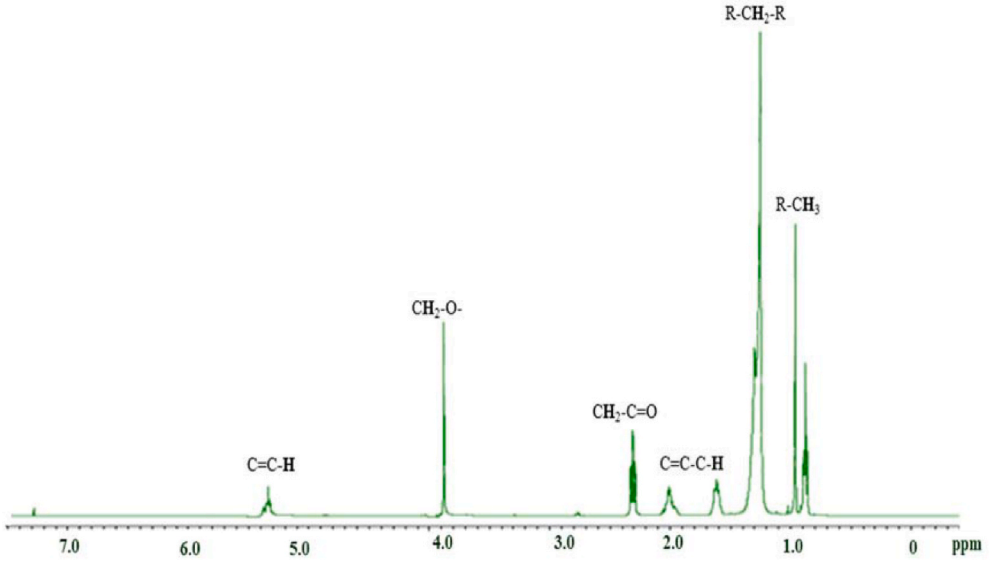


Fig. 7. ^{1}H NMR spectrum of CPOFAs-NPGDE.

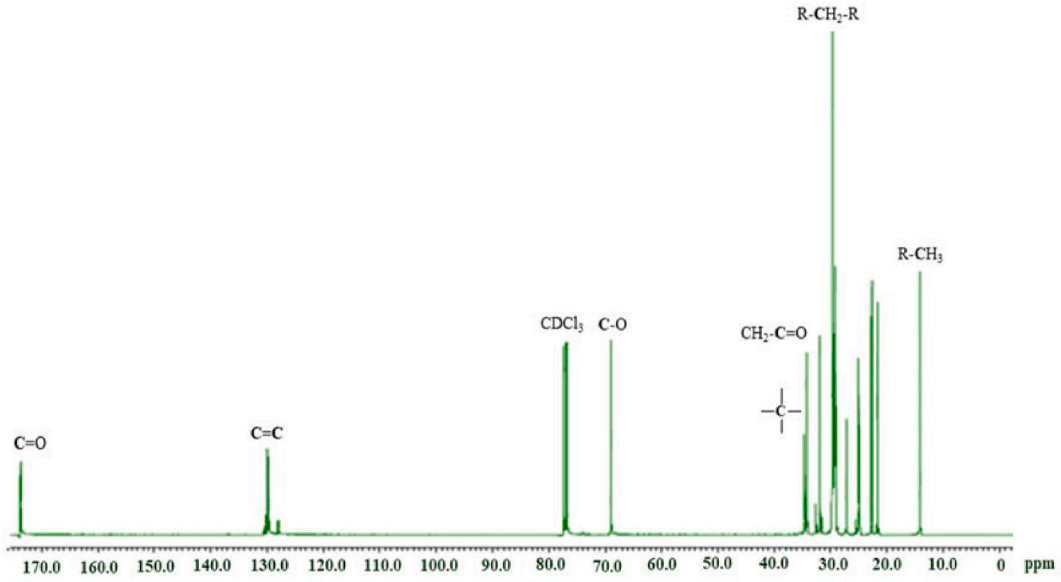


Fig. 8. ^{13}C NMR spectrum of CPOFAs-NPGDE.

carbonyl group (C=O_{acid}) stretching vibrations for CPOFAs showed a sharp peak at 1696, 1291 cm^{-1} of C-O_{acid} and the –OH groups of carboxylic acids (COOH) showed a broad peak at 3304 to 3302 cm^{-1}, respectively [37,41]. After the esterification with polyhydric alcohol (NPG), the polyol ester product, CPOFAs-NPGDE, produced peaks of the ester carbonyl group (C=O_{ester}) and C-O_{ester} for aliphatic esters at 1738 cm^{-1} and 1161 cm^{-1}, respectively. The depletion peaks of hydroxyl (-OH) stretching vibrations and acid carbonyl group (C=O_{acid}) of carboxylic acid proved its conversion to ester product. Complete disappearance of these peaks gives significant evidences of a complete esterification of CPOFAs under the given reaction conditions [40].

One of the most respected techniques for configuration of the chemical structure analysis are ^{1}H and ^{13}C NMR. The success of the esterification reaction conversion can also be certain proven by combination of the FTIR with ^{1}H and ^{13}C NMR analyses results. The ^{1}H and ^{13}C NMR spectra of the final CPOFAs-NPGDE polyol ester are demonstrated in Figs. 7 and 8, respectively. In the ^{1}H NMR spectrum, the proton chemical shifts produced peaks that appeared at: 1.58 ppm for –CH_2-, 1.15–1.24 ppm for –CH_2-, 0.87 ppm for –CH_3, 7.28 ppm for solvent ($CDCl_3$), 3.90 ppm for the RCOO–CH_2– proton of the NPG part; 5.36 ppm and 2.10 ppm for unsaturated aliphatic protons atoms (HC═CH) and methylene group protons (-C═C–C–H). The chemical shift for proton –H_2C–COOH detected at 2.33 for CPOFAs was shifted to 2.27 ppm for the proton of the ester part on carbon α (-H_2C–COOR) and no –COOH at 11.39 ppm [40]. Fig. 8 shows the ^{13}C chemical shifts appeared at: 173.71 ppm for (C═O) ester; 68.96 ppm for (O–C) ester; 35.8 ppm of quaternary carbons (($(CH_3)_2$–C-($CH_2OCOR)_2$ or –C–, 4°); 34.05 ppm for –CH_2-C═O; 130.51 ppm for unsaturated aliphatic carbons atoms (C═C), 32.59, 29.60, 27.19, 24.84 and 22.67 ppm for –CH_2-, 14.08 ppm for terminal methyl (-CH_3) and 77.07 ppm for solvent ($CDCl_3$) [42,43]. Thus, it confirmed the present of polyol ester where CPOFAs-NPGDE diester was successfully formed.

3.9. Polyol ester lubrication properties

The lubrication properties in this study were measured according to the methods with slight adapted from the works of Salih et al. (2012) [42] and Borugadda and Goud (2015) [43]. The lubrication properties such as pour point, flash point, kinematic viscosity, viscosity index, and oxidative stability temperature of the synthesized CPOFAs-NPGDE is

Table 7
Characterization of CPOFAs-NPGDE as biolubricant basestocks

Property	Method	CPOFAs	CPOFAs-NPGDE	Commercial Lube 46A	Commercial Lube 46B	Commercial Lube 46C	Commercial Lube 46D
Iodine value (mg/g)	AOCS Cd 1-25	55 ± 2	45 ± 3	n/a	n/a	n/a	n/a
Density at 25 °C (g/ml)	ASTM D-4052	0.861 ± 0.112	0.878 ± 0.109	0.882 ± 0.115	0.862 ± 0.113	0.890 ± 0.111	0.882 ± 0.112
Viscosity at 40 °C (cSt)	ASTM D-445	52.9 ± 2.5	50.1 ± 3.0	45.7 ± 3.0	46.0 ± 2.5	46.0 ± 4.0	44.2 ± 3.0
Viscosity at 100 °C (cSt)	ASTM D-445	9.2 ± 1.0	15.5 ± 1.0	6.8 ± 1.0	6.7 ± 1.0	6.8 ± 1.0	6.7 ± 1.0
Viscosity index	ASTM -2270	130	190	103	98	101	98
Pour point (°C)	ASTM D-97	10 ± 1	7 ± 1	−23 ± 3	−30 ± 5	−30 ± 4	−27 ± 3
Flash point (°C)	ASTM D-93	240 ± 5	245 ± 4	237 ± 3	213 ± 4	200 ± 3	232 ± 5
Oxidative stability(°C)	ASTM D-6186	181 ± 3	184 ± 3	n/a	n/a	n/a	n/a
ISO Viscosity Grade	ISO 3448	46	46	46	46	46	46
Friction coefficient @ 40 °C @ 100 °C	ASTM D-1894	0.20 0.24	0.190.20	n/a	n/a	n/a	n/a

Lube 46A Denicol Compressor Oil ISO 46.
Lube 46B Maxoline Turbine Oil R&O ISO 46.
Lube 46C PETRONAS Arbor Hydraulic 46.
Lube 46D Mobil DTE™ 20 Series Hydraulic oils.

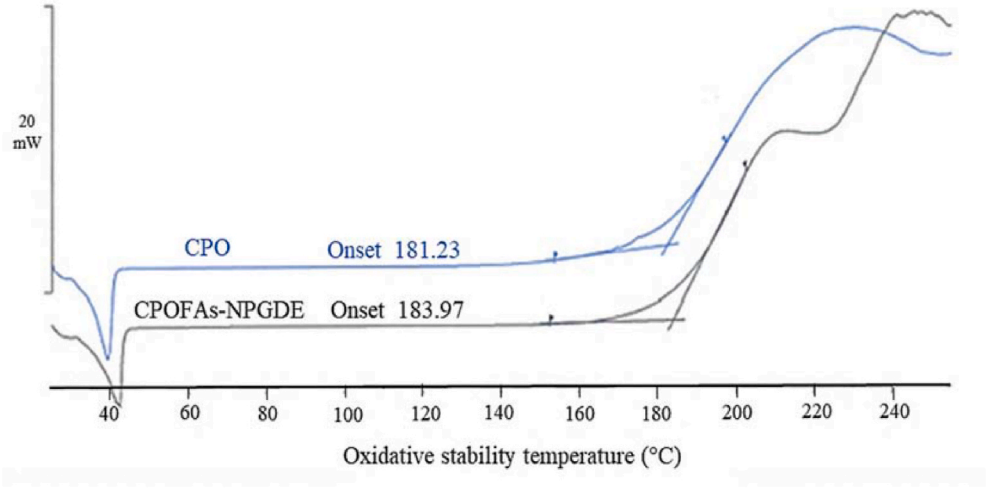


Fig. 9. The PDSC of oxidative stability temperature for the CPO and CPOFAs-NPGDE.

shown in Table 7 and comparation with. The results indicated that the kinematic viscosity values of CPOFAs-NPGDE is 50.1 cSt at 40 °C, and 11.5 cSt at 100 °C. These values would remark the CPOFAs-NPGDE are categories as lubricant ISO VG46 and suitability for use as hydraulic fluids and as fuel oil [44]. The kinematic viscosities of CPOFAs-NPGDE were compared to non-biobased commercial lubricants such as Compressor Oil, Hydraulic Oil, PETRONAS Arbor Hydraulic [45] and Mobil Hydraulic oil with different ISO Viscosity Grade. It was distinguished that the kinematic viscosity values for CPOFAs-NPGDE diester is nearly similar from the commercial lubricants (Compressor Oil, Hydraulic Oil and Maxoline Turbine oil with 46 ISO [46] viscosity grade). The lubricants viscosity index highpoints how the viscosity changes through the variations in temperature. A high viscosity index designates that the viscosity is basically constant over a wide range of the temperature [47]. The resultant CPOFAs-NPGDE diester from this study has shown high viscosity index value at 190 where it can be classified as high viscosity index fluids base esters [48]. In general, the viscosity index value of the CPOFAs-NPFDE is highest than commercial lubricants such as Denicol Compressor Oil, Castrol and Subs Tech Hydraulic Oil, PETRONAS Arbor Hydraulic and Maxoline Turbine oil.

A potential biolubricant can also be evaluated from its pour point, which is when the liquid can still be poured. For engines in cold climates, the biolubricant should have a low pour point. Generally, lower pour points have been observed for polyol esters with more short-chain saturated fatty acids (C_4–C_{12}) and more unsaturated fatty acids [49]. In this study, the pour point of the synthesized CPOFAs-NPGDE was 7 °C and higher than pour points of the selected commercial lubricants. This indicates CPOFAs-NPGDE suitable to be used as biolubricant in tropical climate countries. The international standard of pour point for paraffin distillates based commercial lubricants were in the range of −7 °C to −18 °C [50]. Therefore, the synthesized CPOFAs-NPFDE is as good as the commercial lubricants where pour point depressants need to be added if lower pour points oils are required at low temperature application.

Flash point is the indicator of the volatility and flammability of biolubricant [51]. In this study, the flash point value of the CPOFAs-NPGDE was compared with commercial lubricants such as Denicol Compressor Oil ISO 46, Petronas Arbor Hydraulic 46 Oil, and Mobil Hydraulic with the same ISO 46 Viscosity Grade. The results show that the CPOFAs-NPGDE had highest flash points (245 ± 4 °C) as compared to other non-biobased commercial lubricants as shown in Table 7. This finding showed that some bio-based lubricant such as CPOFAs-NPGDE has higher flash point as compared to petroleum-based lubricants. However most international lubricating standards require petroleum-based lubricants with high flash point at 290 °C while ester-based lubricants at 250 °C [44,52], respectively.

Further performance of polyol ester as biolubricants must shows good oxidative stability temperature is essential to be considered [44]. An effective instrumentation for measuring the oxidative stability temperature (OS_T) in an accelerated mode is pressure differential scanning calorimetry (PDSC) [53]. The OS_T is the temperature where a rapid upsurge in the rate of oxidation is observed. A high value of OS_T would suggest that the substance could resilient to oxidation process at particular temperature [54,55]. The capability of a substance to resist oxidative degradation is significant characteristic for biolubricants. Therefore, CPO as raw material and CPOFAs-NPGDE polyol ester were screened for their oxidation stability temperature (OS_T). The OS_T was calculated from a plot of heat flow (W/g) versus temperature for each tested CPO before esterification and CPOFAs-NPGDE. Fig. 9 shows the oxidative stability comparison for CPO before esterification was 181 °C, whereas for CPOFAs-NPGDE was increased to 184 °C. Based on the results the synthesized CPOFAs-NPGDE has acceptable OS_T values compared to others palm-based polyol ester such as CPO-TMP (183 °C) [35], CPKO-di-TMPTE (199.7 °C) [44], CPKO-TMPTE (245 °C) [56] and PAO6 (OS_T, 201 °C) and would be suitable to be used as a commercial biolubricant. However, CPOFAs-NPGDE has lower OS_T as compared CPKO based polyol ester such as CPKO-TMPTE (triester) and CPKO-di-TMPTE (tetraester). This is due to more higher polyol degree of TMP (tri-) and di-TMP (tetra-) esters for CPKO-based polyol ester having more molecular weight and thus having more oxidative stability temperature [56]. As overall, the synthesized CPOFAs-NPGDE in this study is compatible with some of the commercialized lubricants. Based on ISO VG 46 lubricant grade, COFAs-NPGDE is potentially to be used as hydraulic oil, compressor and turbine oil system.

3.10. Tribological properties

The tribological performances of the biolubricant are dominated through the interactions of the biolubricant fluid with metal surfaces. The performance of biolubricants especially under boundary conditions highly depend on their chemical molecular structures. The occurrence of an important ester polar head groups and nonpolar straight-chain carbon tail groups in the synthesized ester molecular structure made it amphiphilic in nature, therefore allowing it to be used as boundary lubricants. The ester head groups polarity favour to adsorb on a metallic surface [57], which decreases the surface energy [42] and causes a reduction of the friction coefficient (FC) [53]. Therefore, the polarity of the biolubricant increases its efficiency in reducing wear [27]. The results in this work indicated that the minimum FC for the hydrodynamic regime of CPOFAs-NPGDE at 40 °C was 0.19 whereas, the minimum FC

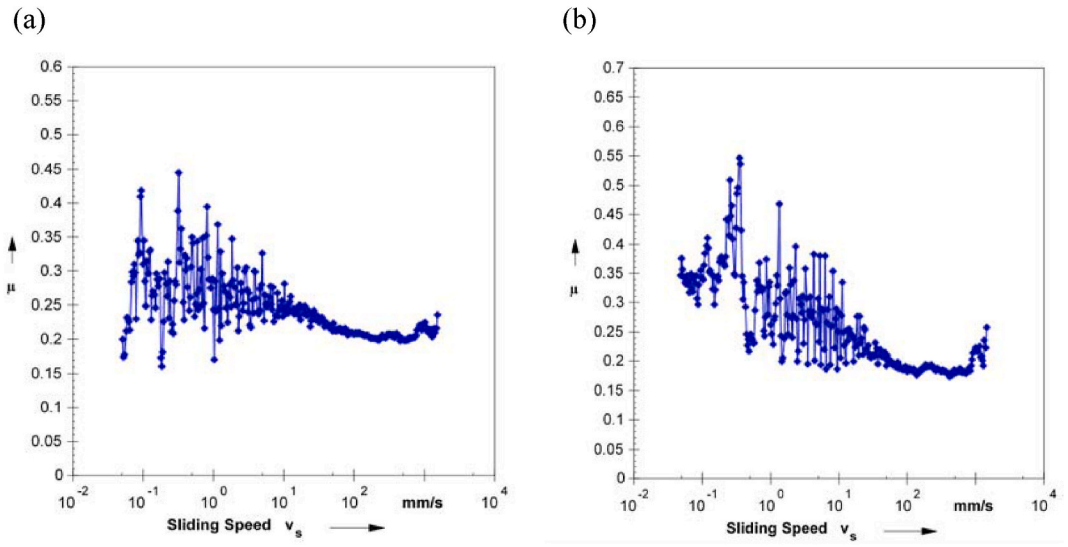


Fig. 10. Stribeck curve plot for CPOFAs-NPGDE.

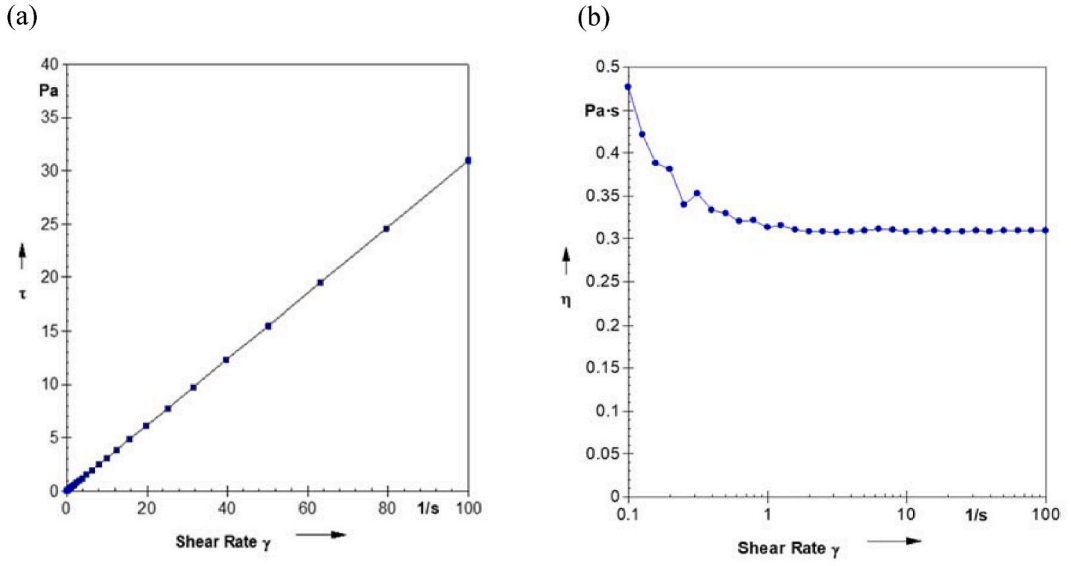


Fig. 11. Shear stress *vs* shear rate plot (a) and viscosity vs shear rate plot (b) of CPOFAs-NPGDE.

of CPOFAs-NPGDE at 100 °C was 0.20 (Fig. 10). It showed low coefficient of friction values and comparative to some dicarboxylate esters [58] and USFA-TMP ester [59]. It is also at par as compared to of Petronas Arbor Hydraulic 46 and Maxoline Turbine Oil R&O ISO 46 lubricants. The ester molecular structure of CPOFAs-NPGDE with high polarity has high tendency to form unimolecular barrier adsorbed protective layer on the metal surfaces thus produces low energy absorption and produce low friction. This indicates that better protective layers have been formed and thus reduce the FC values. The biolubricant produced, CPOFAs-NPGDE recorded a low friction coefficient below 0.5 at 40 °C and 100 °C as recommended for commercial lubricants [57]. These demonstrations the good quality of tribological properties even in high temperatures and is able to compete with commercial lubricants in the market.

3.11. *Rheological properties*

Rheological Newtonian properties of the CPOFAs-NPGDE was identified through the relationship between shear stress and shear rate. When the relationship is linear, the fluid is classified as Newtonian [57, 60]. The classification, either as Newtonian or non-Newtonian also depends on their viscosity by increasing shear rate. Constant viscosity is considered as Newtonian, while inconstant viscosity is non-Newtonian [61]. Based on Fig. 11, it is known that the relation between shear stress and shear rate is linear which would classify CPOFAs-NPGDE as a Newtonian fluid. This indicates that the viscosity of CPOFAs-NPGDE will not change as the shear stress increases. This was supported by a constant viscosity against shear rate at 10-100 s^{-1}.

4. Conclusions

The esterification process of CPOFAs with NPG was successfully carried out to obtain the polyol ester-based biolubricant, CPOFAs-NPGDE. The process responses of polyol ester yield and diester selectivity were improved through the optimization using response surface methodology by D-optimal design. The optimal condition for the esterification process was successfully obtained at CPOFAs:NPG molar ratio of 2.26:1, 1.12% H_2SO_4 catalyst, reaction temperature of 138 °C for

4.79 h reaction time and produced 87.6% yield with 100% diester composition of CPOFAs-NPGDE. The polyol ester product, CPOFAs-NPGDE showed good lubrication properties that include low pour point (10 °C) and friction coefficient (0.19–0.20), high values of flash-point (235 °C), viscosity index (190), and good OS_T (184 °C), which were compatible with the commercial lubricants with the same ISO viscosity grade. CPOFAs-NPGDE is classify as a Newtonian fluid bio-lubricant. Therefore, it has the potential to be used in various industrial applications such as hydraulic, compressor and turbine fluids.

CRediT authorship contribution statement

Nurazira Mohd Nor: Investigation, Methodology, SoftwareSofware. **Nadia Salih:** Visualization, Writing – original draft, Writing – review & editing. **Jumat Salimon:** Conceptualization, Funding acquisition, Resources, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.renene.2022.09.112.

References

[1] Z. Zakaria, A.R.A. Rahim, Z. Aman, Issues and challenges of oil palm cooperatives towards greater sustainability: a proposal of conceptual framework, Int. J. Acad. Res. Bus. Soc. Sci. 10 (1) (2020) 46–69, https://doi.org/10.6007/IJARBSS/v10-i1/6806.

[2] W.A. Ahmed, N. Salih, J. Salimon, Lubricity, tribological and rheological properties of green ester oil prepared from bio-based azelaic acid, Asian J. Chem. 33 (2021) 1363–1369, https://doi.org/10.14233/ajchem.2021.23209.

[3] P. Gutbrod, S. Reichert, K. Gutbrod, A. Hamai, C. Bréhélin, G. Ngando-Ebongue, P. Dörmann, Fatty acid isoprenoid alcohol ester synthesis in fruits of the African oil palm (*Elaeis guineensis*), Phytochemistry (Oxf.) 185 (2021), 112684, https://doi.org/10.1016/j.phytochem.2021.112684.

[4] M. Bahadi, N. Salih, J. Salimon, Synthesis and characterization of green biodegradable palm oleic acid based polyester, Biointerface Res. Appl. Chem. 11 (2021) 14359–14371, https://doi.org/10.33263/BRIAC116.1435914371.

[5] MPOB, Economics and industry development devision. http://bepi.mpob.gov.my/index.php/en/?option=com_content&view=category&id=106, 2021. (Accessed 1 June 2022). Accessed.

[6] F.N. Maluin, M.Z. Hussein, A.S. Idris, An overview of the oil palm industry: challenges and some emerging opportunities for nanotechnology development, Agronomy 10 (2020) 356, https://doi.org/10.3390/agronomy10030356.

[7] N. Salih, J. Salimon, A Review on eco-friendly green biolubricants from renewable and sustainable plant oil sources, Biointerface Res. Appl. Chem. 11 (2021) 13303–13327, https://doi.org/10.33263/BRIAC115.1330313327.

[8] Global biolubricants markets report/industry report, 2022 – 2026 global market trajectory & analytics by research and markets 2022, increasing focus on biodegradable, non-toxic, and eco-friendly products, Markets Report 2022-2026, https://www.globenewswire.com/en/news-release/2022/04/06/2417401/28124/en/Global-Biolubricants-Markets-Report-2022-2026-Increasing-Focus-on-Biodegradable-Non-toxic-and-Eco-Friendly-Products.html. (Accessed 1 June 2022). Accessed.

[9] Y.N. Lye, N. Salih, J. Salimon, Optimization of partial epoxidation on *Jatropha curcas* oil based methyl linoleate using urea-hydrogen peroxide and methyltrioxorhenium catalyst, Appl. Sci. Eng. Prog. 14 (2021) 88–99, https://doi.org/10.14416/j.asep.2020.12.006.

[10] S. qing Wu, T. ting Sun, Z. zhe Cai, J. Shen, W. zhe Yang, Z. min Zhao, D. po Yang, Biolubricant base stock with improved low temperature performance: ester complex production using housefly (Musca domestica L.) larval lipid, Renew. Energy 162 (2020) 1940–1951, https://doi.org/10.1016/j.renene.2020.10.001.

[11] N. Salih, J. Salimon, A Review on new trends, challenges and prospects of ecofriendly friendly green food-grade biolubricants, Biointerface Res. Appl. Chem. 12 (2022) 1185–1207, https://doi.org/10.33263/BRIAC121.11851207.

[12] S. Samidin, N. Salih, J. Salimon, Synthesis and characterization of trimethylolpropane based esters as green biolubricant basestock, Biointerface Res. Appl. Chem. 11 (2021) 13638–13651, https://doi.org/10.33263/BRIAC115.1363813651.

[13] D.J. Murphy, K. Goggin, R.R.M. Paterson, Oil palm in the 2020s and beyond: challenges and solutions, CABI Agric. Biosci. 2 (2021) 39, https://doi.org/10.1186/s43170-021-00058-3.

[14] A. Alfieri, E. Imperlini, E. Nigro, D. Vitucci, S. Orrù, A. Daniele, P. Buono, A. Mancini, Effects of plant oil interesterified triacylglycerols on lipemia and human health, Int. J. Mol. Sci. 19 (2018) 104, https://doi.org/10.3390/ijms19010104.

[15] H. Kim, N. Choi, Y. Kim, H.R. Kim, J. Lee, I.H. Kim, Immobilized lipase-catalyzed esterification for synthesis of trimethylolpropane triester as a biolubricant, Renew. Energy 130 (2019) 489–494, https://doi.org/10.1016/j.renene.2018.06.092.

[16] M.A. Jumaah, F.L. Khaleel, N. Salih, J. Salimon, Synthesis of Tri, Tetra, and Hexa-Palmitate Polyol Esters from Malaysian Saturated Palm Fatty Acid Distillate for Biolubricant Production, Biomass Conv. Bioref., 2022, https://doi.org/10.1007/s13399-022-02517-x. Article in Press.

[17] S.-q. Wu, Z.-z. Cai, Y. Niu, D. Zheng, G. r He, Y. Wang, D. p Yang, A renewable lipid source for biolubricant feedstock oil from housefly (*Musca domestica*) larva, Renew. Energy 113 (2017) 546–553, https://doi.org/10.1016/j.renene.2017.05.094.

[18] J. Calero, D. Luna, E.D. Sancho, C. Luna, F.M. Bautista, A.A. Romero, A. Posadillo, J. Berbel, C. Verdugo-Escamilla, An overview on glycerol-free processes for the production of renewable liquid biofuels, applicable in diesel engines, Renew. Sustain. Energy Rev. 42 (2015) 1437–1452, https://doi.org/10.1016/j.rser.2014.11.007.

[19] N.H. Arbain, J. Salimon, N. Salih, W.A. Ahmed, Optimization for epoxidation of Malaysian *Jatropha curcas* oil based trimethylolpropane ester biolubricant, Appl. Sci. Eng. Prog. (2021), https://doi.org/10.14416/j.asep.2021.10.009. Article in Press.

[20] R.N.M. Kamil, S. Yusup, Modeling of reaction kinetics for transesterification of palm-based methyl esters with trimethylolpropane, Bioresour. Technol. 101 (2010) 5877–5884, https://doi.org/10.1016/j.biortech.2010.02.084.

[21] A.N. Annisa, W. Widayat, A Review of bio-lubricant production from vegetable oils using esterification transesterification process, MATEC Web Conf 156 (2018) 1–7, https://doi.org/10.1051/matecconf/201815606007.

[22] D. Kania, R. Yunus, R. Omar, S.A. Rashid, B.M. Jan, A review of biolubricants in drilling fluids: recent research, performance and applications, J. Pet. Sci. Eng. 135 (2015) 177–184, https://doi.org/10.1016/j.petrol.2015.09.021.

[23] É.C.G. Aguieiras, E.D.C. Cavalcanti, P.R. da Silva, V.F. Soares, R. Fernandez-Lafuente, C.L.B. Assunção, J.A.C. da Silva, D.M.G. Freire, Enzymatic synthesis of neopentyl glycol-bases biolubricants using biodiesel from soybean and castor bean as raw materials, Renew. Energy 148 (2020) 689–696, https://doi.org/10.1016/j.renene.2019.10.156.

[24] K.V. Fernandes, E.D.C. Cavalcanti, E.P. Cipolatti, E.C.G. Aguieiras, M.C.C. Pinto, F. A. Tavares, P.R. da Silva, R. Fernandez-Lafuente, S. Arana-Peña, J.C. Pinto, C.L. B. Assunção, J.A.C. da Silva, D.M.G. Freire, Enzymatic synthesis of biolubricants from by-product of soybean oil processing catalyzed by different biocatalysts of *Candida rugosa* lipase, Catal. Today 362 (2021) 122–129, https://doi.org/10.1016/j.cattod.2020.03.060.

[25] A. Papadaki, K.V. Fernandes, A. Chatzifragkou, E.C.G. Aguieiras, J.A.C. da Silva, R. Fernandez-Lafuente, S. Papanikolaou, A. Koutinas, D.M.G. Freire, Bioprocess development for biolubricant production using microbial oil derived via fermentation from confectionery industry wastes, Bioresour. Technol. 267 (2018) 311–318, https://doi.org/10.1016/j.biortech.2018.07.016.

[26] N.A. Raof, R. Yunus, U. Rashid, N. Azis, Z. Yaakub, Palm-based neopentyl glycol diester: a potential green insulating oil, Protein Pept. Lett. 25 (2018) 171–179, https://doi.org/10.2174/0929866525666180122095056.

[27] M.A. Jumaah, N. Salih, J. Salimon, Optimization for esterification of saturated palm fatty acid distillate by D-optimal design response surface methodology for biolubricant production, Turk. J. Chem. 45 (2021) 1391–1407, https://doi.org/10.3906/kim-2103-11.

[28] R.N.M. Kamil, S. Yusup, L. Ismail, Production by esterification of neopentyl glycol with hexanoic acid, Asian J. Chem. 25 (2013) 8256–8260, https://doi.org/10.14233/ajchem.2013.14362.

[29] V. Rajaendran, J. Salimon, R.M. Yusop, Synthesis and characterization of epoxidized neopenthyl glycol diolate as an intermediate of biolubricant, Malaysian J. Anal. Sci. 20 (2016) 1329–1337, https://doi.org/10.17576/MJAS-2016-2006-11.

[30] The European Chemicals Agency, ECHA registered dossier for fatty acids, C12-16, esters with neopentyl glycol - Registration. https://echa.europa.eu/registration-dossier/-/registered-dossier/8041/6/2/1. (Accessed 1 June 2022). Accessed.

[31] U. Resit, A.L. Roger, L.M. Mark, Response surface model Building and multidisciplinary optimization using D-optimal designs, in: 7th AIAA/USAF/NASA/ISSMO Symposium on Multidisciplinary Analysis and Optimization, 1998, https://doi.org/10.2514/6.1998-4759.

[32] R. Holm, I.H.M. Jensen, J. Sonnergaard, Optimization of self-microemulsifying drug delivery systems (SMEDDS) using a D-Optimal design and the desirability function, Drug Dev. Ind. Pharm. 32 (2006) 1025–1032, https://doi.org/10.1080/03639040600559024.

[33] E. Kuram, B. Ozcelik, M. Bayramoglu, E. Demirbas, B.T. Simsek, Optimization of cutting fluids and cutting parameters during end milling by using D-optimal design of experiments, J. Clean. Prod. 42 (2013) 159–166, https://doi.org/10.1016/j.jclepro.2012.11.003.
[34] M. Bahadi, N. Salih, J. Salimon, D-Optimal design optimization for the separation of oleic acid from Malaysian high free fatty acid crude palm oil fatty acids mixture using urea complex fractionation, Appl. Sci. Eng. Prog. 14 (2021) 175–186, https://doi.org/10.14416/j.asep.2021.03.004.
[35] N.M. Nor, D. Derawi, J. Salimon, Esterification and evaluation of palm oil as biolubricant base stock, Malaysian J. Chem. 21 (2019) 28–35.
[36] H.A. Mamud, N. Salih, J. Salimon, Oleic acid based polyolesters of TMP and PE for biolubricant application, Malaysian J. Anal. Sci. 19 (2015) 97–105.
[37] N.M. Nor, N. Salih, J. Salimon, Optimization of the ring opening of epoxidized palm oil using D-optimal design, Asian J. Chem. 33 (2021) 67–75, https://doi.org/10.14233/ajchem.2021.22938.
[38] I. Grčić, D. Vujević, N. Koprivanac, The use of D-optimal design to model the effects of process parameters on mineralization and discoloration kinetics of Fenton-type oxidation, Chem. Eng. J. 157 (2010) 408–419, https://doi.org/10.1016/j.cej.2009.11.030.
[39] A.A. Japir, N. Salih, J. Salimon, Synthesis and characterization of biodegradable palm palmitic acid based bioplastic, Turk. J. Chem. 45 (2021) 585–599, https://doi.org/10.3906/kim-2011-31.
[40] M.A. Bahadi, A.-W. Japir, N. Salih, J. Salimon, Free fatty acids separation from Malaysian high free fatty acid crude palm oil using molecular distillation, Malaysian J. Anal. Sci. 20 (2016) 1042–1051, https://doi.org/10.17576/mjas-2016-2005-08.
[41] D.L. Pavia, G.M. Lampman, G.S. Kriz, J.R. Vyvyan, Introduction to Spectroscopy, fifth ed., Cengage Learning, Inc., United States, 2015.
[42] N. Salih, J. Salimon, E. Yousif, The physicochemical and tribological properties of oleic acid based triester biolubricants, Ind. Crop. Prod. 34 (2011) 1089–1096, https://doi.org/10.1016/j.indcrop.2011.03.025.
[43] V.B. Borugadda, V.V. Goud, Response surface methodology for optimization of biolubricant basestock synthesis from high free fatty acids castor oil, Energy Sci. Eng. 3 (2015) 371–383, https://doi.org/10.1002/ese3.77.
[44] M. Bahadi, J. Salimon, D. Derawi, Synthesis of di-trimethylolpropane tetraester-based biolubricant from Elaeis guineensis kernel oil via homogeneous acid-catalyzed trans- esterification, Renew. Energy 171 (2021) 981–993, https://doi.org/10.1016/j.renene. 2021.02.163.
[45] Petronas arbor hydraulic 46 oil. https://www.pli-petronas.com/en/agriculture-and-construction/products/petronas-arbor-hydraulic-46-hv-iso-vg-46. (Accessed 1 June 2022). Accessed.
[46] Maxoline turbine oil R&O ISO 46 lubricants oil. http://www.maxoline.com/our-products/product-listing/item/turbine-oil-r-o-iso-vg-46. (Accessed 1 June 2022). Accessed.
[47] N. Razali, H. Mootabadi, B. Salamatinia, K.T. Lee, A.Z. Abdullah, Optimization of process parameters for alkaline-catalysed transesterification of palm oil using response surface methodology, Sains Malays. 39 (2010) 805–809.
[48] M.A. Jumaah, N. Salih, J. Salimon, D-Optimal design optimization for esterification of palm fatty acids distillate with polyhydric alcohols for biolubricants production, Iran. J. Chem. Chem. Eng. (Int. Engl. Ed.) (2021), https://doi.org/10.30492/IJCCE.2021.521586.4481. Article in Press.
[49] S.K. Tulashie, F. Kotoka, The potential of castor, palm kernel, and coconut oils as biolubricant base oil via chemical modification and formulation, Therm. Sci. Eng. Prog. 16 (2020), 100480, https://doi.org/10.1016/j.tsep.2020.100480.
[50] B. Angulo, J.M. Fraile, L. Gil, C.I. Herrerías, Bio-lubricants production from fish oil residue by transesterification with trimethylolpropane, J. Clean. Prod. 202 (2018) 81–87, https://doi.org/10.1016/j.jclepro.2018.07.260.
[51] L.A. Quinchia, M.A. Delgado, C. Valencia, J.M. Franco, C. Gallegos, Natural and synthetic antioxidant additives for improving the performance of new biolubricant formulations, J. Agric. Food Chem. 59 (2011) 12917–12924, https://doi.org/10.1021/jf2035737.
[52] N.M. Nor, N. Salih, J. Salimon, Chemically modified *Jatropha curcas* oil for biolubricant applications, Hem. Ind. 75 (2021) 117–128, https://doi.org/10.2298/HEMIND200809009N.
[53] I.M. Alarifi, A.B. Alkouh, V. Ali, H.M. Nguyen, A. Asadi, On the rheological properties of MWCNT-TiO_2/oil hybrid nanofluid: an experimental investigation on the effects of shear rate, temperature, and solid concentration of nanoparticles, Powder Technol. 355 (2019) 157–162, https://doi.org/10.1016/j.powtec.2019.07.039.
[54] ASTM, Standard Test Method for Wear Preventive Characteristics of Lubricating Fluid (Four-ball Method), ASTM, USA, West Conshohocken, PA, 2011.
[55] M.A. Jumaah, N. Salih, J. Salimon, Optimization process for esterification of Malaysian palm fatty acids distillate with high degree polyhydric alcohols, Bioenerg. Res. (2022), https://doi.org/10.1007/s12155-022-10415-3. Article in Press.
[56] M. Bahadi, M.F. Yusoff, J. Salimon, D. Derawi, Optimization of response surface methodology by D-optimal design for synthesis of food-grade palm kernel based biolubricant, Ind. Crop. Prod. 139 (2019), 111452, https://doi.org/10.1016/j.indcrop.2019.06.015.
[57] J. Salimon, W.A. Ahmed, N. Salih, M.A. Yarmo, D. Derawi, Lubricity and tribological properties of dicarboxylic acid and 2-butyl-1- octanol based ester, Sains Malays. 44 (2015) 405–412.
[58] W. Abdo Ahmed, N. Salih, J. Salimon, Synthesis, characterization, tribological and rheological properties of di (2-butyloctyl) dicarboxylate esters for environmentally friendly biolubricant applications, Biointerface Res. Appl. Chem. 13 (3) (2023) 278, https://doi.org/10.33263/BRIAC133.278.
[59] M. Ahmed Jumaah, N. Salih, J. Salimon, D-Optimal Design Optimization of Unsaturated Palm Fatty Acid Distillate and Trimethylolpropane Esterification for Biolubricant Production, in: Comptes Rendus. Chimie, Online first, 2022, pp. 1–20, https://doi.org/10.5802/crchim.178.
[60] N. Al-Arafi, N. Salih, J. Salimon, Synthesis, characterization, tribological and rheological properties of oleyl oleate based biolubricant. Egypt, J. Chem. 65 (2022) 419–433, https://doi.org/10.21608/ejchem.2021.77715.3797.
[61] P. Coussot, L. Tocquer, C. Lanos, G. Ovarlez, Macroscopic vs. local rheology of yield stress fluids, J. Non-Newtonian Fluid Mech. 158 (2009) 85–90, https://doi.org/10.1016/j.jnnfm.2008.08.003.